# Analysis of Battery Bidding Behaviours Through Classification Algorithms in UK Power Markets

Yujie Wu
Energy Institute, University College London

**Abstract**

This study applies machine learning classification algorithms to analyse bidding behaviours of battery energy storage systems (BESS) across multiple UK electricity markets during 2024. Using comprehensive data from Elexon and NESO covering four market segments—including Enduring Auction Capability (EAC) services, Balancing Mechanism, Balancing Reserve, and wholesale markets—the research investigates whether battery operators possess distinct strategic "fingerprints" that differentiate their market participation approaches.

Random Forest and Naive Bayes algorithms were trained to classify battery operators based on their normalized bidding patterns across 48 half-hourly settlement periods. The analysis encompasses bidding data from major battery operators including Tesla Motors, EDF Energy Customers, Statkraft Markets, Limejump Energy, Flexitricity, and others, representing the comprehensive behavioural analysis of UK battery storage market participation. Decision path clustering analysis also reveals tactical sub-patterns within strategic groups.

These findings provide empirical evidence that UK battery storage markets support strategic diversity. The research also establishes a methodological framework for behavioural analysis of BESS in electricity markets, contributing to understanding of how market design influences strategic differentiation among energy storage operators.

**Keywords:** battery energy storage systems; bidding behaviour; UK power markets; machine learning classification; ancillary services

## 1 Summary

Batteries are playing an increasingly pivotal role in electricity markets worldwide. The global deployment of battery energy storage has surged dramatically in recent years—for example, the United States has seen its installed battery storage capacity surge from around 2.4 GW in 2020 to exceeding 30 GW by early 2025, marking over a tenfold increase [1]. This rapid growth underscores the rising importance of battery energy storage systems (BESS) in enabling a more flexible and low-carbon grid. These systems operate by storing electricity in chemical form and later releasing it when needed, essentially acting as a buffer that absorbs surplus generation and discharges during peak demand. Through this capability, batteries help balance supply and demand in real time and facilitate the integration of intermittent renewable resources into the grid [2].

The UK serves as an illustrative case of a fast-evolving battery market where companies must adapt quickly. There are multiple revenue opportunities for batteries in the UK—from the Balancing Mechanism (BM) and ancillary frequency response services to reserve markets like Short-Term Operating Reserve (STOR) and the wholesale energy market—and operators often engage in "revenue stacking" by participating across several of these markets [3]. In recent years the UK grid has introduced new ancillary services (for example, a Quick Reserve product launched in late 2024 alongside a Balancing Reserve), which significantly boosted revenues for battery units able to provide rapid response [4]. At the same time, increasing competition among storage providers has started to compress profit margins in established services—a phenomenon known as price cannibalization [5]. As a result, battery companies in Britain are frequently recalibrating how they bid into each market segment to remain competitive and capture the best value from their assets.

This research utilizes comprehensive bidding data sourced from two key UK electricity market organizations: Elexon and the National Energy System Operator (NESO). Elexon serves as the Settlement Administration Agent for the UK electricity market, and NESO manages the transmission network and procures ancillary services. From publicly available datasets, we collected Balancing Mechanism Unit (BMU) information, detailed bid-offer data from the Balancing Mechanism, Physical Notification, and EAC sell orders. While detailed wholesale electricity market data and capacity market information remain commercially sensitive and are not publicly disclosed, Physical Notification data are employed as a proxy indicator for wholesale market trading positions.

To the best of knowledge, this analysis represents the first comprehensive study of its kind to apply machine learning classification techniques to battery storage bidding behaviours in the UK market, providing novel insights into the strategic bidding patterns of major battery operators using publicly available market data.

Given this dynamic context, three key research questions are investigated:

1. Can machine learning classifiers (specifically Random Forest algorithms) successfully identify battery operators from their daily bidding patterns, which indicate that companies possess distinct strategic approaches to market participation?

2. What distinctive patterns in bidding strategies and market participation behaviours can be identified for different battery operators?

3. How do different battery operators stack their bidding across multiple markets, and do the identified bidding behaviours correspond to distinct market preferences or varied approaches within the same markets?

This research applies classification algorithms to bidding data from multiple market arenas for a set of large battery operators. By training models to identify which company submitted a given bid, this study tests whether each firm has a unique strategic "fingerprint" or whether their bidding patterns demonstrate convergence.

# 2 Materials and Methods

## 2.1 Data Collection

This study analyses battery storage bidding behaviours across multiple UK electricity markets during the period 2024/01/01 to 2024/12/31. All volume-based bidding behaviours are measured in MW/hour, with normalisation method mentioned in next subsection.

Data were collected from publicly available APIs provided by NESO and ELEXON. The data collection strategy focused on capturing bidding decisions rather than revenue outcomes. Table 1 summarises the data sources and key columns used.

Table 1: Data collection overview

| Market | Data Source | Used Columns | Column Explanations |
|---|---|---|---|
| EAC | NESO Data Portal | deliveryStart, deliveryEnd, auctionUnit, serviceType, auctionProduct, quantity | Service delivery time windows, BMU identifier, service classification, volume (MW) |
| Balancing Mechanism | ELEXON BMRS Bid-Offer | settlementDate, settlementPeriod, nationalGridBmUnit, levelFrom, levelTo | Trading day and 30-min period, BMU identifier, MW capacity range |
| Wholesale | ELEXON BMRS PN | settlementDate, settlementPeriod, levelFrom, levelTo, nationalGridBmUnit | Planned generation/consumption range (MW) |
| Balancing Reserve | NESO Balancing-Reserve Sell Orders | auctionID, auctionUnit, basketID, serviceType, deliveryStart, deliveryEnd, quantity | Time windows, BMU identifier, MW volume |
| BMU Data | ELEXON BMRS BMU LIST | nationalGridBmUnit, leadPartyName, fuelType, demandCapacity, generationCapacity | Company/operator name, technology classification, technical limits (MW) |

### 2.1.1 Data Cleaning

All bidding quantities were processed as MW/hour for each settlement period. For EAC data with different temporal structures, volumes were converted to half-hourly equivalents labelled by settlement periods. Battery assets were identified using the `fuelType` field in Elexon's BMU list, filtering for units marked as `BATTERY`. However, some battery installations may appear under `OTHER` fuel type categories; notable examples include Jamesfield 1, Jamesfield 2, Little Raith, and Roaring Hill, which were included after manual verification.

### 2.1.2 Data Exclusions

The Capacity Market was excluded due to lack of publicly available detailed bidding data. Imbalance/cash-out settlements were excluded as they represent settlement outcomes rather than strategic bidding. STOR data were collected but formed a very small dataset with minimal impact. Wholesale market analysis relies on Physical Notifications as a proxy, following the methodology developed by Modo Energy [6].

## 2.2 Bidding Behaviour Data Generation

The longitudinal bidding data were transformed into a cross-sectional format where each row represents a single instance (day and BMU) and the 48 settlement periods serve as feature columns, containing normalised bid volumes. Table 2 shows an example.

Table 2: Example of cross-sectional data format for bidding behaviours.

| deliveryDate | auctionUnit | participant | 1 | 2 | ... | 47 | 48 |
|---|---|---|---|---|---|---|---|
| 2024-01-01 | AG-GSTK05 | STATKRAFT MARKETS GMBH | 0.0 | 0.0 | ... | 0.216 | 0.216 |
| 2024-01-01 | AG-HLIM04 | LIMEJUMP ENERGY LIMITED | 0.0 | 0.0 | ... | -0.037 | -0.037 |
| ... | ... | ... | ... | ... | ... | ... | ... |

### 2.2.1 EAC Services

Volume extraction uses the `quantity` column for each delivery window. The `auctionProduct` identifier categorises products into high frequency services (DCH, DMH, DRH, NQR: absorb energy) and low frequency services (DCL, DML, DRL, PQR: inject energy). For baskets containing parent and child orders, quantities are summed.

### 2.2.2 Balancing Mechanism

Data preprocessing excludes rows where both `levelFrom` and `levelTo` equal zero. Volume is calculated as the average MW: (levelFrom + levelTo)/2. Following BM conventions, bids are associated with reducing output, offers with increasing output.

### 2.2.3 Balancing Reserve

Similar to EAC, quantities are summed across parent-child basket structures. Positive Balancing Reserve (PBR) quantities are divided by generation capacity and Negative Balancing Reserve (NBR) by demand capacity for normalisation.

### 2.2.4 Wholesale Market (Physical Notification)

Processing mirrors the Balancing Mechanism methodology, calculating average MW as (levelFrom + levelTo)/2 for each settlement period.

## 2.3 Data Normalisation

Raw volumes are normalised using BMU capacity data to focus on bidding strategy rather than asset scale:

$$\text{Normalised Volume} = \frac{\text{Raw Volume (MW/h)}}{\text{BMU Capacity (MW/h)}}.$$

Capacity reference data come from Elexon's BMU dataset, representing generation capacity and demand capacity. The distributions of these capacities are shown in Figures 1 and 2.

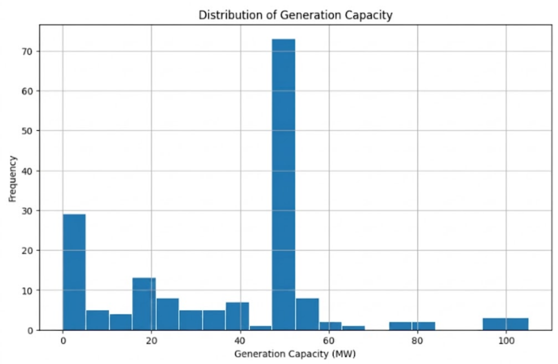


Figure 1: Distribution of generation capacity of BMU.

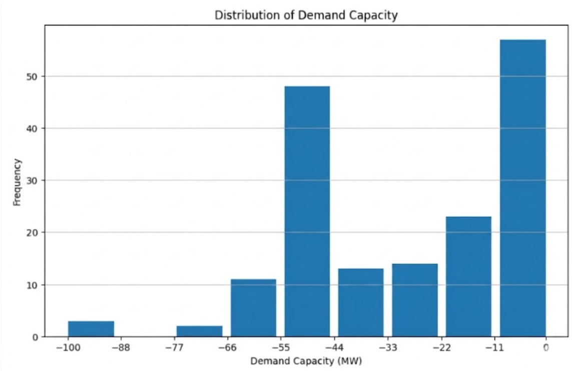


Figure 2: Distribution of demand capacity of BMU.

## 2.4 Data Exploratory Analysis

Before building classification models, we performed an exploratory analysis to understand the overall participation patterns, temporal distributions, and correlations across market segments. The distributions of normalised bidding volumes for EAC services are shown together in Figure 3. Low- and high-frequency products are plotted in paired panels, so DCL and DCH, DML and DMH, DRL and DRH, and PQR and NQR can be compared directly within the same graph. Quick Reserve services (NQR, PQR) and Dynamic Moderation (DMH, DML) display highly concentrated distributions with most bids clustered at small volumes. Dynamic Regulation shows stronger asymmetry between high and low products, while Dynamic Containment (DCH, DCL) distributions are broader, indicating more flexible capacity allocation.

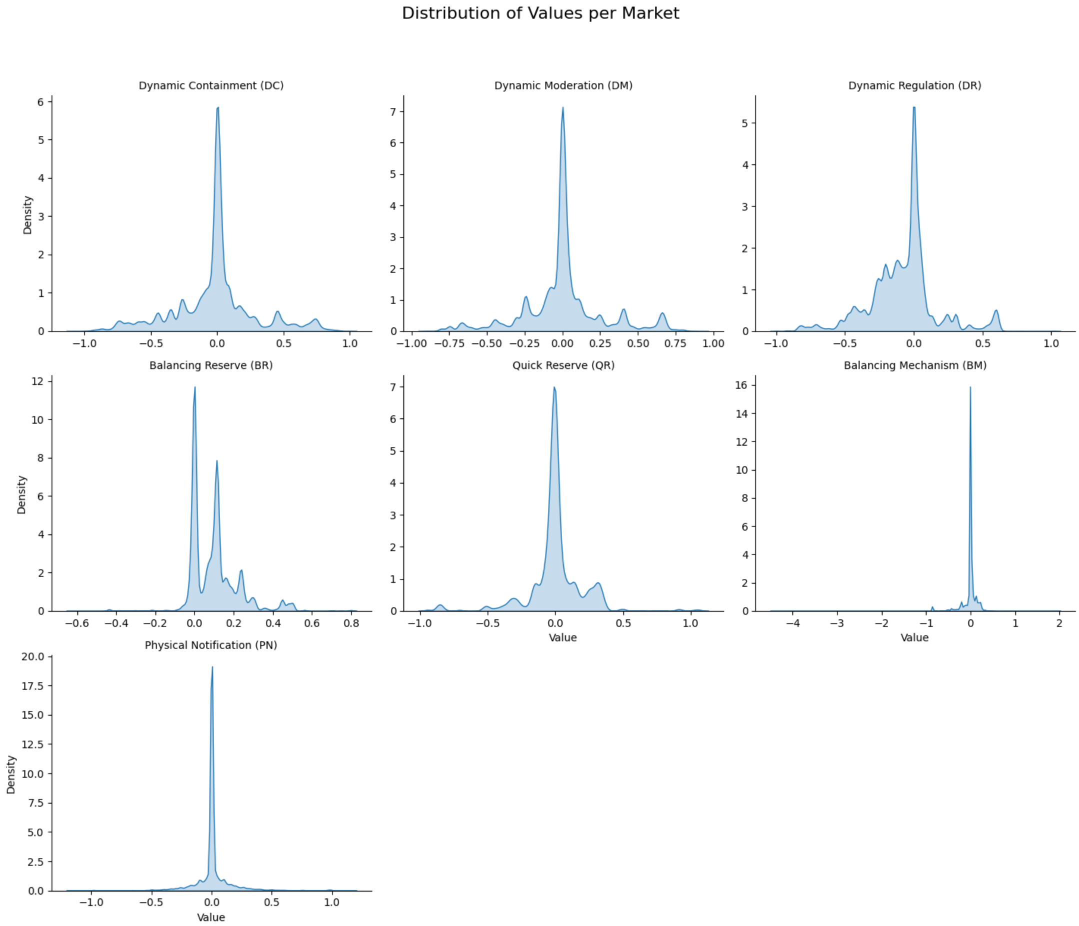


Figure 3: Normalised EAC bidding volume distributions by markets.

### 2.4.1 EAC Market Participation

Figure 4 shows the order counts per month across EAC auction products throughout 2024. A dramatic spike in Positive Quick Reserve (PQR) and Negative Quick Reserve (NQR) is observed in December 2024, indicating immediate adoption of the newly launched Quick Reserve service.

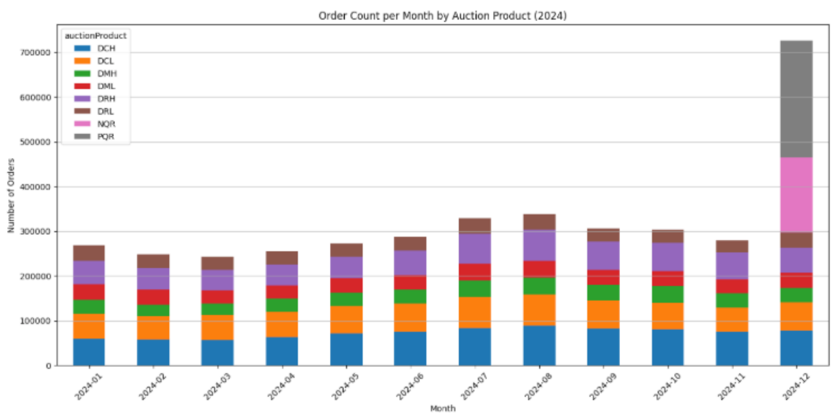


Figure 4: Monthly bidding order numbers by auction product in EAC.

The correlation heatmap of EAC bidding volumes (Figure 5) reveals strong positive correlations between complementary direction pairs, such as DCH–DCL and NQR–PQR, suggesting that operators bid similar volumes for positive and negative directions within the same service. Strong negative correlations appear between Dynamic Regulation and Dynamic Containment, reflecting different operational roles.

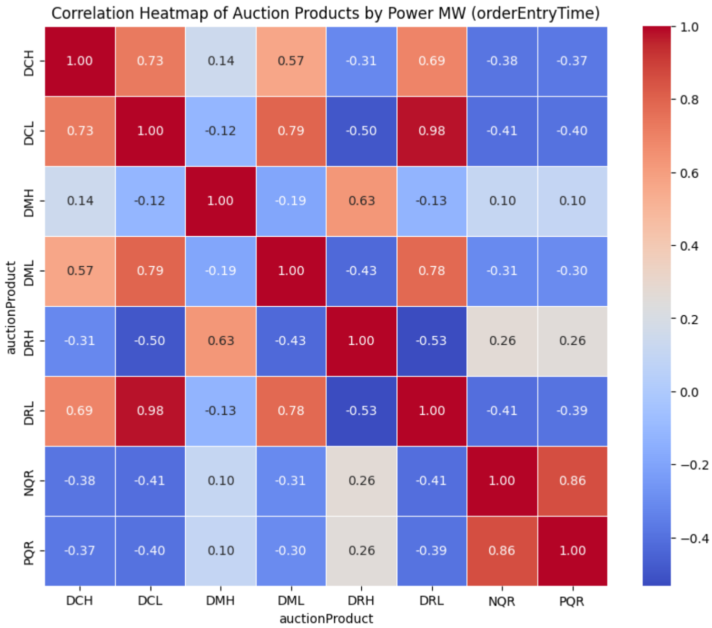


Figure 5: Correlation matrix of bidding volumes by EAC auction product.

The below figure visualises three types of batteries bidding window in Dynamic Containment High Service (Figure 6). Most batteries take exact the mid-day window to charge, while some avoid the most competitive bidding time window. This image also provide a method to identify auction windows of each market.

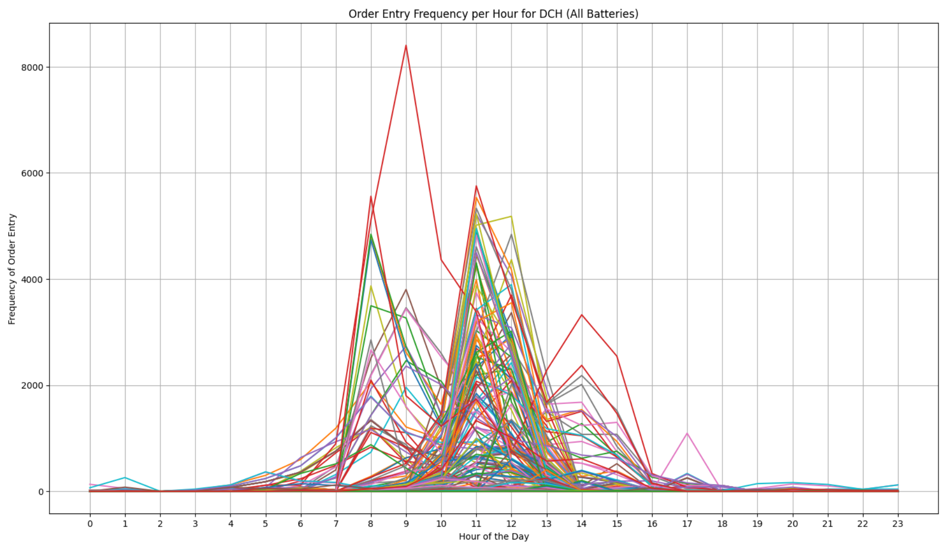


Figure 6: Total order number by settlement period in DCH of all batteries.

Among the major participants, Tesla leads dynamic-service bidding, EDF is the strongest broad follower, Statkraft maintains significant presence in DC.Figure 7

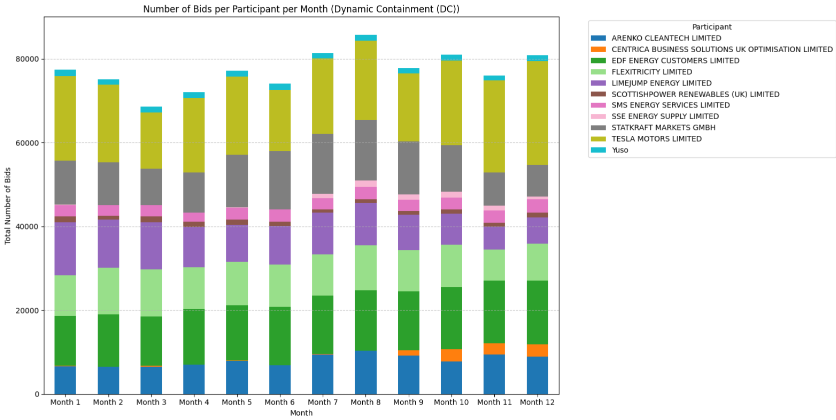


Figure 7: Dynamic Containment operators' participation by month.

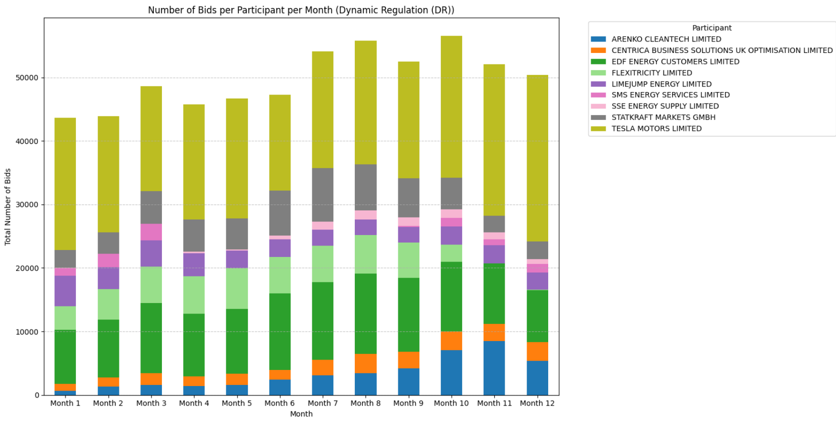


Figure 8: Dynamic Regulation operators' participation by month.

### 2.4.2 Balancing Mechanism Participation

Balancing Mechanism participant dominance is stable across the year. Statkraft remains the largest contributor in most months, followed by EDF, Tesla, and Limejump.(Figures 9)

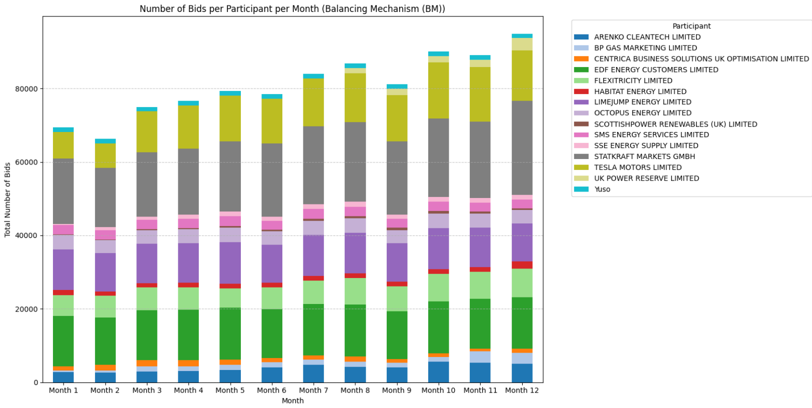


Figure 9: BM Bid and offer participation by month.

Bid and offer count breakdowns for selected operators(Figures 10) show that some batteries (e.g., EDF) are generation-oriented, while others exhibit more balanced or demand-side participation.

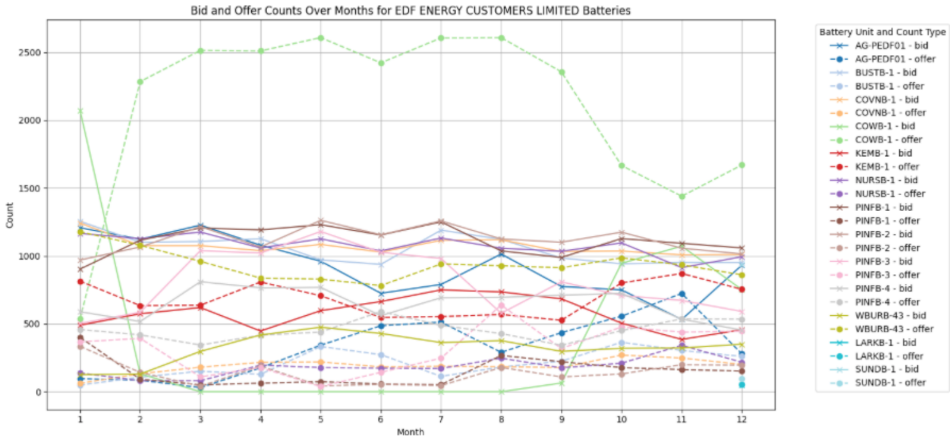


Figure 10: BM Bid and offer counts per month for EDF Customers Limited batteries.

The correlation matrix of BM decisions across settlement periods (Figure 11) reveals that bidding behaviours cluster by operational time blocks rather than only by adjacent periods: morning periods are highly correlated with other morning periods, and evening periods are correlated with each other. A noticeable correlation also appears between noon and midnight/overnight periods.

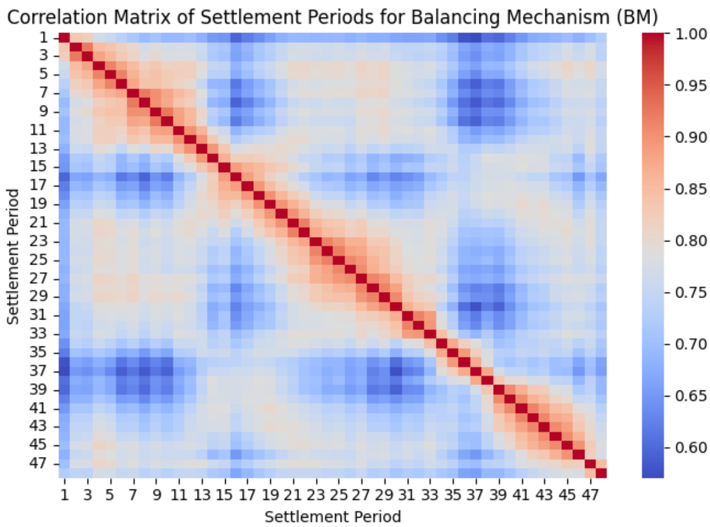


Figure 11: Balancing Mechanism settlement period correlation matrix.

### 2.4.3 Balancing Reserve Participation

Balancing Reserve (BR) volume distribution shows a strong central peak at zero with secondary peaks at 0.2 and 0.4 normalised volume (Figure 3), suggesting that certain operators consistently bid at specific fractional capacities. Positive Balancing Reserve (PBR) dominating (Figure 3). Tesla appears the major operator (Figure 12).

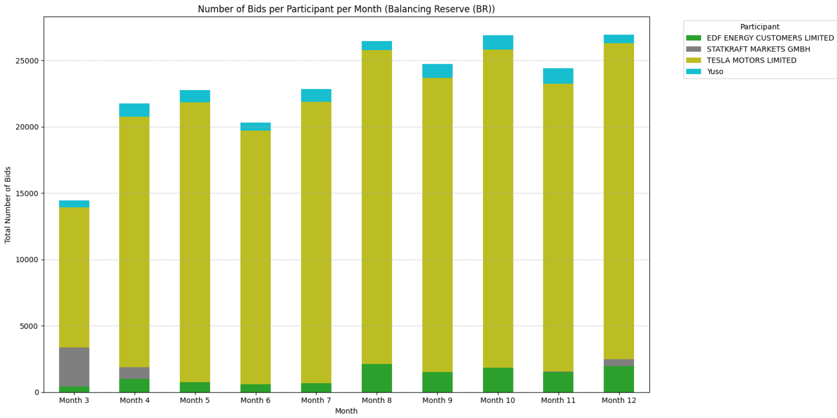


Figure 12: BR operators' participation by month.

### 2.4.4 Wholesale Market (Physical Notification)

Physical Notifications (PN), used as a proxy for wholesale trading, exhibit a near-normal distribution centred around zero with a slight positive skew (Figure 3), indicating that batteries more frequently position for export (discharge) than import. Table 13 shows strong PN participation across all major operators.

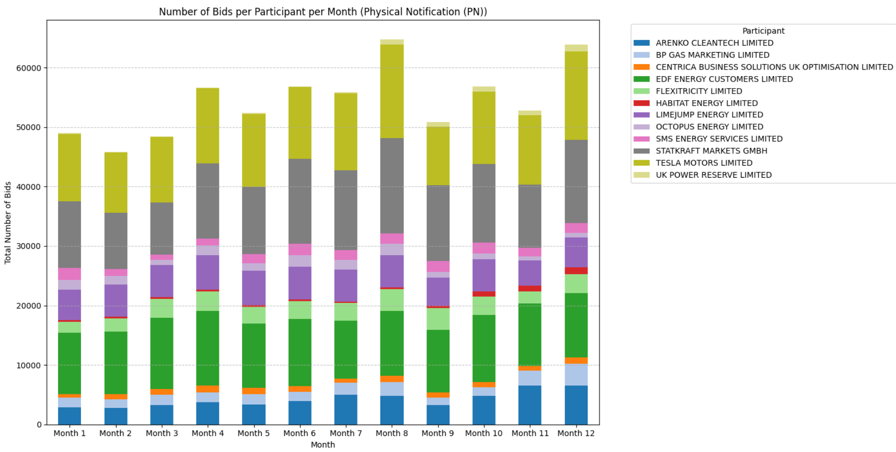


Figure 13: PN operators' participation by month.

### 2.4.5 Overall Market Participation

Combining all markets, we observe that Dynamic Containment and Dynamic Moderation tend to be more positive during early/midday blocks and more negative around the evening peak (especially SP31–44). Dynamic Regulation is mostly negative across the day, with occasional positive blocks overnight or late day, suggesting it is often used for charging or state-of-charge management. Physical Notification is frequently negative in the morning, strongly positive around SP35–40, and moderately positive near midday, consistent with evening peak discharge and intraday repositioning. Quick Reserve only appears in Month 12 and has a strong structured profile: negative at the start/end of the day, positive across most daytime periods. Balancing Mechanism is mostly mildly negative across all months and periods, appearing as a persistent charging or counter-positioning layer.

Figure 14 (a composite visualisation) summarises the average normalised volume per market and settlement period across all operators. This multi-market view confirms that different services fulfil distinct roles in the operators' revenue-stacking strategies.

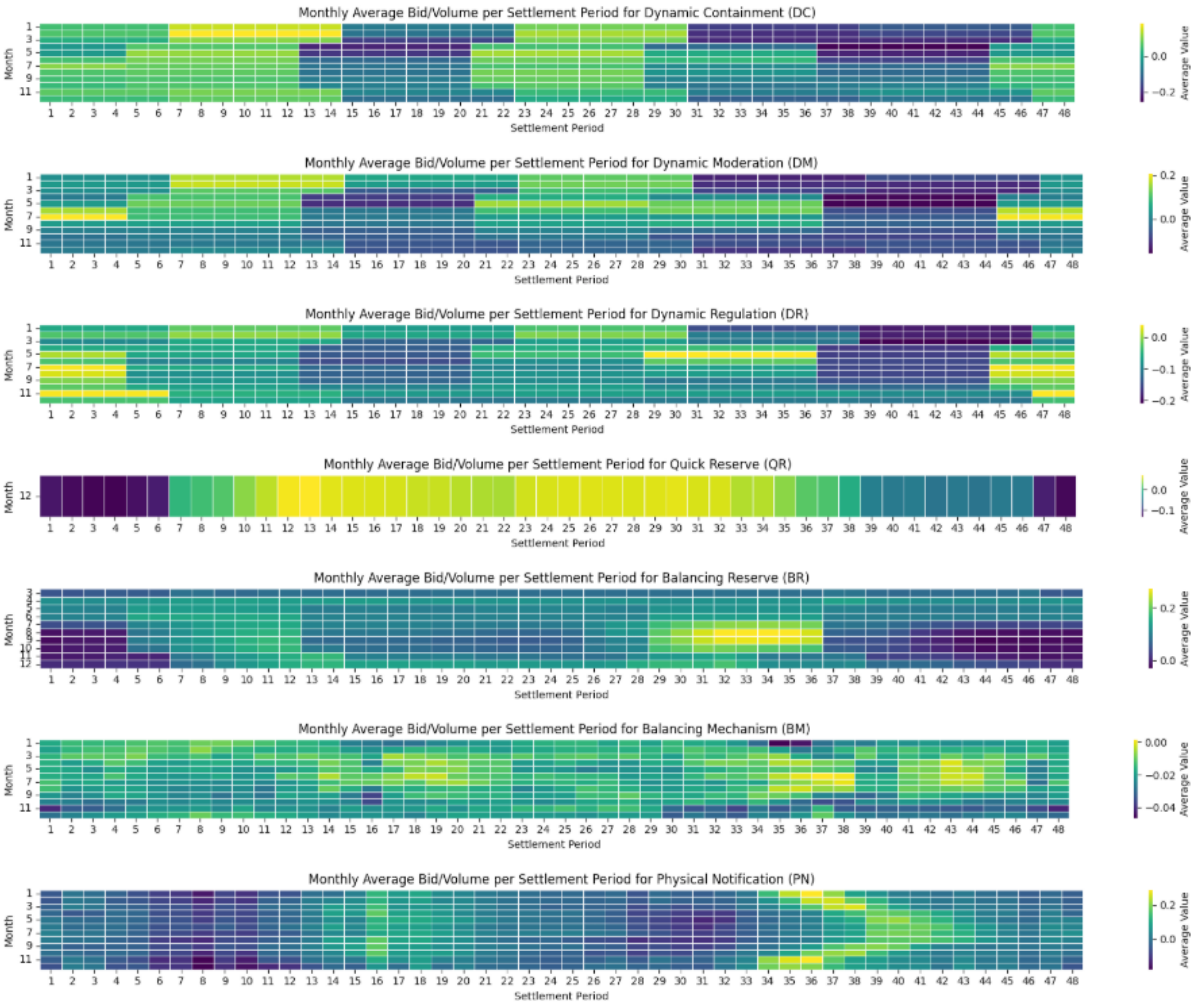


Figure 14: Overall market participation profiles: average normalised volume per settlement period across all operators.

## 2.5 Classification Approach

Two machine learning algorithms were compared: Naive Bayes as a baseline and Random Forest as the primary classifier. Naive Bayes assumes feature independence, providing a benchmark. Random Forest captures non-linear feature interactions and handles high-dimensional data without requiring explicit independence assumptions.

## 2.6 Evaluation Approach

Given severe class imbalance in the UK battery market, model performance is evaluated using Overall Accuracy, Weighted F1-score, and Macro F1-score. The Macro F1-score is the primary metric as it treats all market participants equally, validating the algorithm's ability to capture diverse strategic fingerprints of smaller operators.

# 3 Results

## 3.1 Classification Report

Table 3 summarises the performance of both classifiers across the seven market segments. Random Forest substantially outperforms Naive Bayes in all markets, indicating that bidding behaviours involve complex feature interactions.

Table 3: Classification performance summary (Accuracy, Macro F1, Weighted F1)

| | Random Forest | | | Naive Bayes | | |
|---|---|---|---|---|---|---|
| Market | Acc | MF1 | WF1 | Acc | MF1 | WF1 |
| Dynamic Containment | 0.87 | 0.78 | 0.87 | 0.29 | 0.24 | 0.29 |
| Dynamic Moderation | 0.91 | 0.79 | 0.90 | 0.29 | 0.22 | 0.30 |
| Dynamic Regulation | 0.95 | 0.93 | 0.95 | 0.41 | 0.32 | 0.44 |
| Quick Reserve | 0.94 | 0.87 | 0.94 | 0.85 | 0.72 | 0.86 |
| Balancing Reserve | 0.97 | 0.69 | 0.97 | 0.67 | 0.38 | 0.74 |
| Balancing Mechanism | 0.77 | 0.64 | 0.76 | 0.30 | 0.24 | 0.32 |
| Physical Notification | 0.67 | 0.48 | 0.64 | 0.20 | 0.15 | 0.17 |

## 3.2 Naive Bayes Baseline

Confusion matrix examples. The Naive Bayes confusion matrices (Figures 15, 16 and 17) reveal poor performance, with most predictions dominated by the majority class.

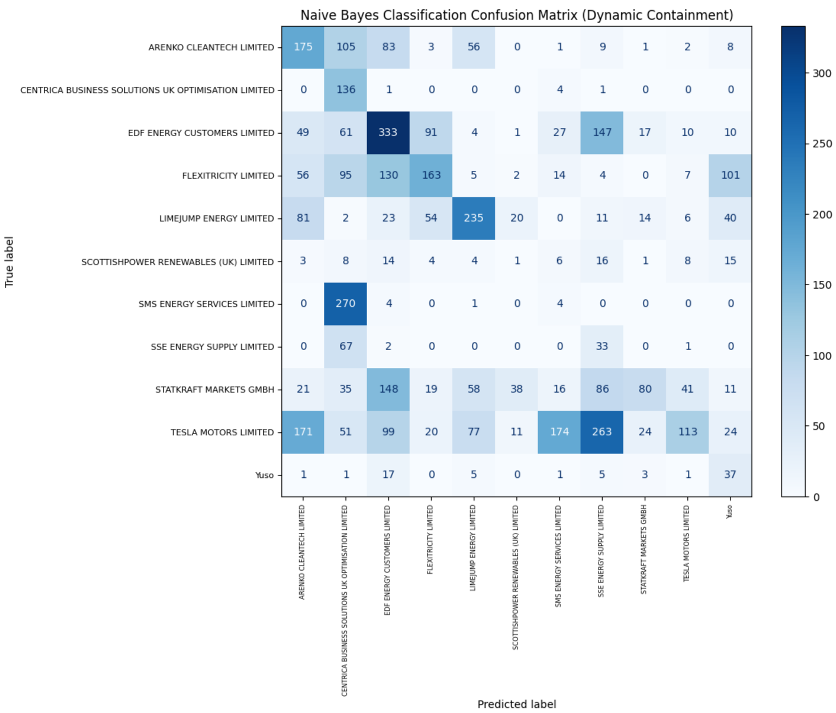


Figure 15: Naive Bayes classification confusion matrix – Dynamic Containment.

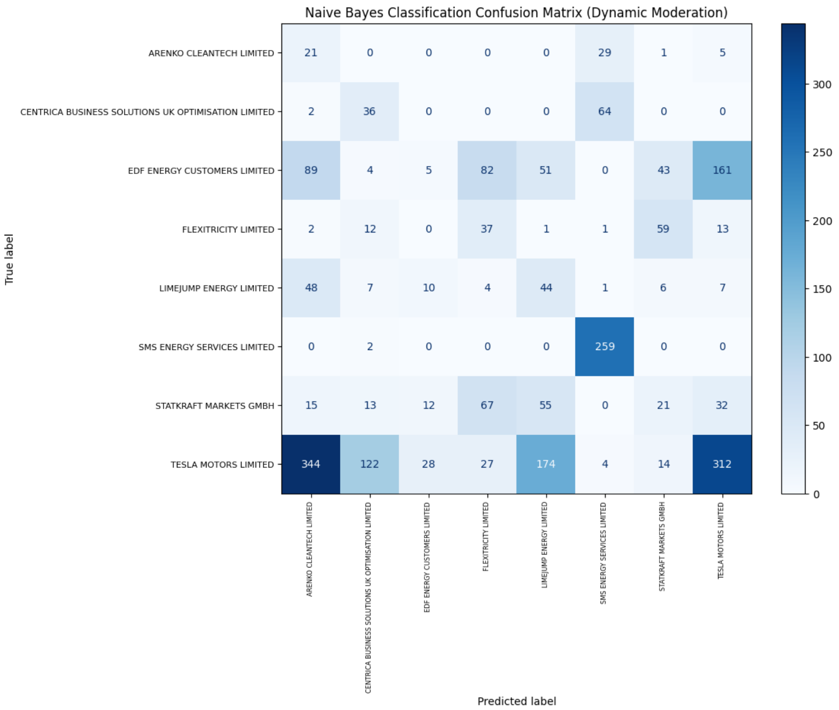


Figure 16: Naive Bayes classification confusion matrix – Dynamic Moderation.

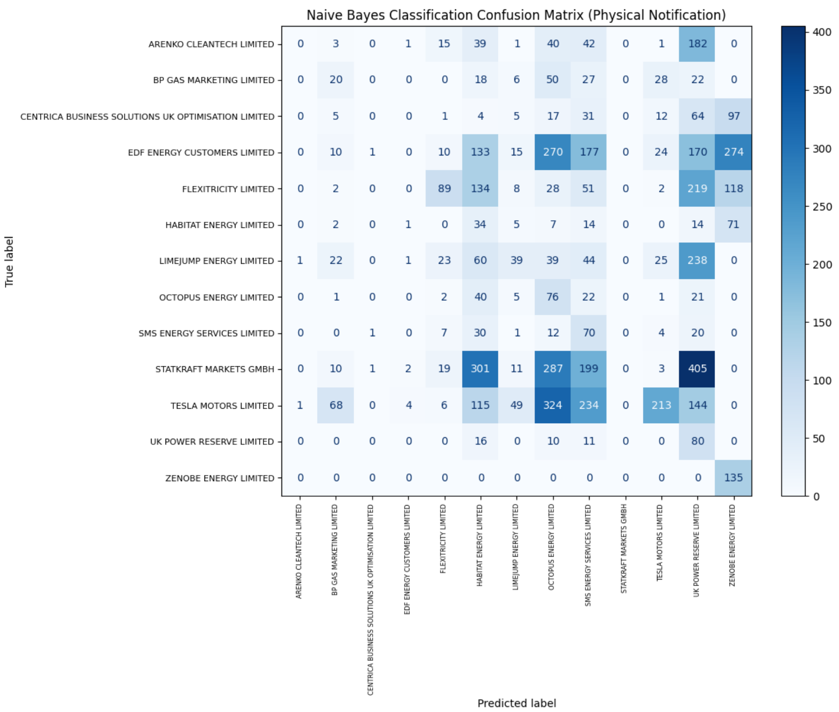


Figure 17: Naive Bayes classification confusion matrix – Physical Notification.

## 3.3 Random Forest Performance

Confusion matrix examples. Random Forest confusion matrices (Figures 18, 19 and 20) show much stronger diagonal dominance, demonstrating successful identification of operator-specific strategies.

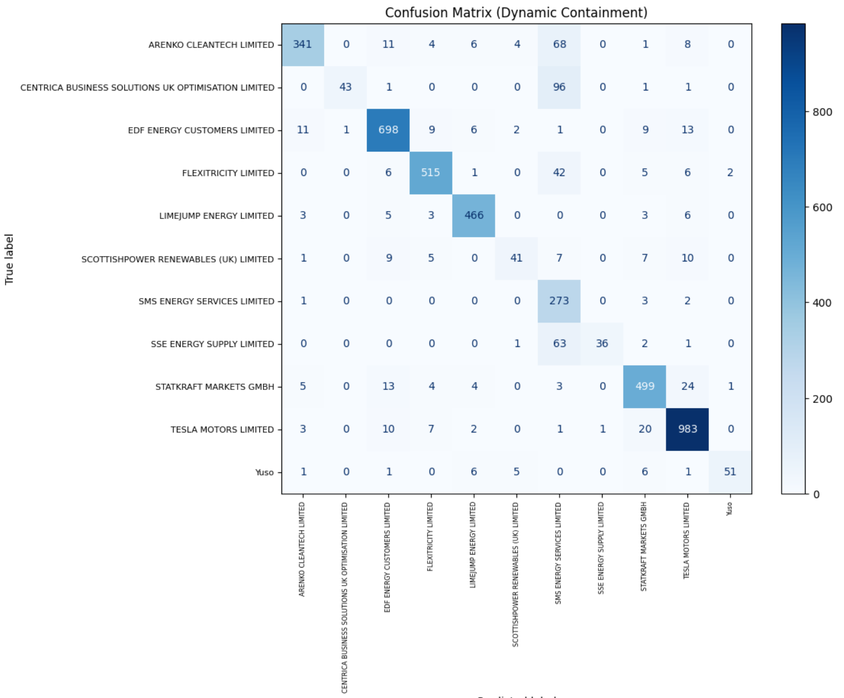


Figure 18: Random Forest classification confusion matrix – Dynamic Containment.


Figure 19: Random Forest classification confusion matrix – Dynamic Moderation.

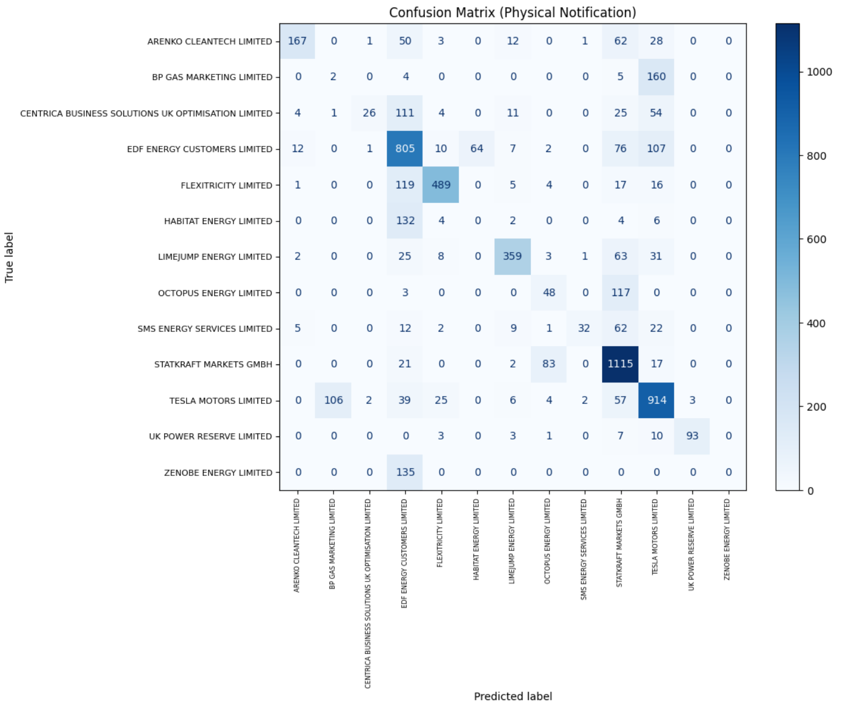


Figure 20: Random Forest classification confusion matrix – Physical Notification.

## 3.4 Random Forest Feature Importance

Feature importance analysis (Figures 21–27) reveals distinct temporal patterns that differentiate operator strategies. Quick Reserve and Dynamic Regulation show extreme concentration in period 48, while Balancing Mechanism discrimination occurs primarily in early periods (1–3). Several services concentrate discriminative power during traditional peak demand periods (17–22, 35–45), might suggesting that strategic differentiation is most pronounced when market values are highest.

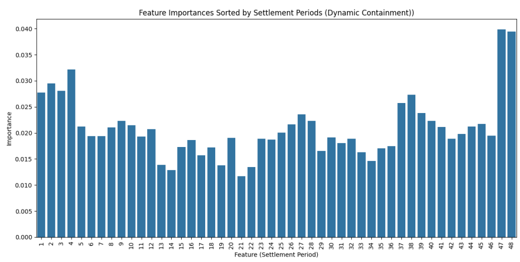


Figure 21: Dynamic Containment.

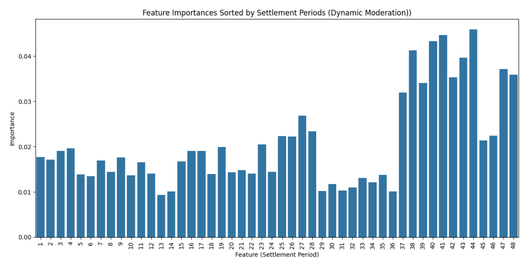


Figure 22: Dynamic Moderation.

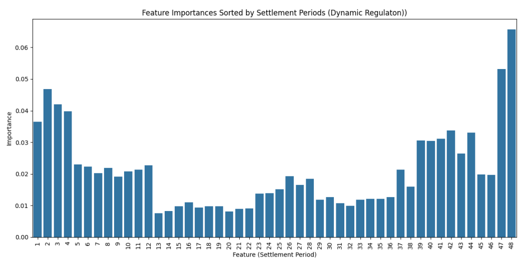


Figure 23: Dynamic Regulation.

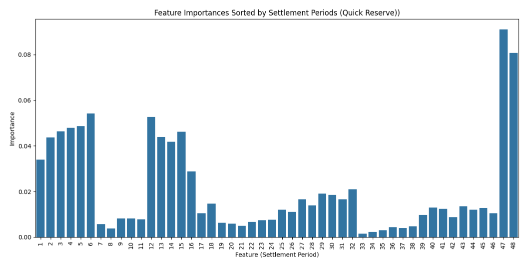


Figure 24: Quick Reserve.

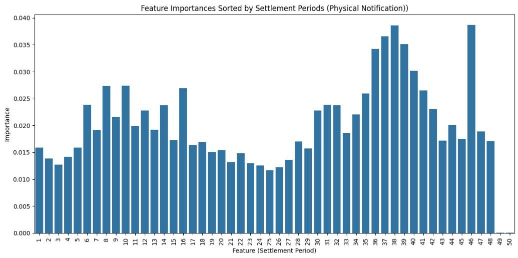


Figure 25: Physical Notification.

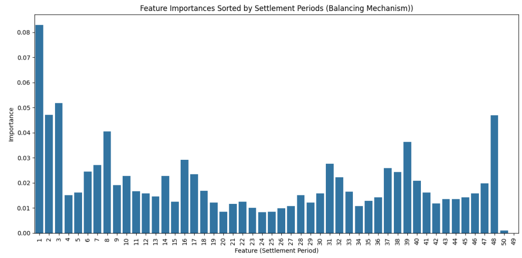


Figure 26: Balancing Mechanism.

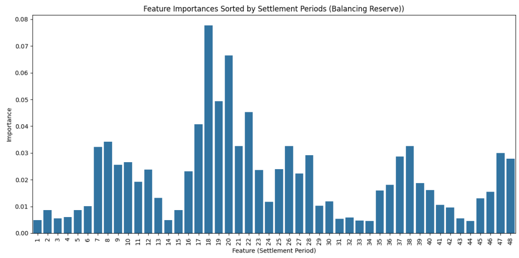


Figure 27: Balancing Reserve.

## 3.5 Pattern Clustering

Decision path clustering was applied within each operator group to identify tactical sub-patterns. The method was chosen from k-means, DBSCAN, and spectral clustering. Clusters were evaluated using silhouette score, and the best clustering solution was selected. The Cramér's V association between cluster membership and auction unit/month was computed to assess whether strategies are asset-specific or temporally driven. But the limitation of small sample size and mixed effect of variables should be awared.

Detailed bidding patterns for each market are presented in Figures 28–39 and discussed below.

### 3.5.1 Dynamic Containment

DC patterns align with Electricity Forward Agreement (EFA) blocks. Two primary charging windows appear: morning (07:00–11:00, EFA block 3) and afternoon/evening (15:00–22:00, blocks 5 and 6). Most companies simultaneously bid for high and low frequency services using loop basket structures. Flexitricity might demonstrates asset-specific strategies (Cramér's V = 0.87 with auction unit).

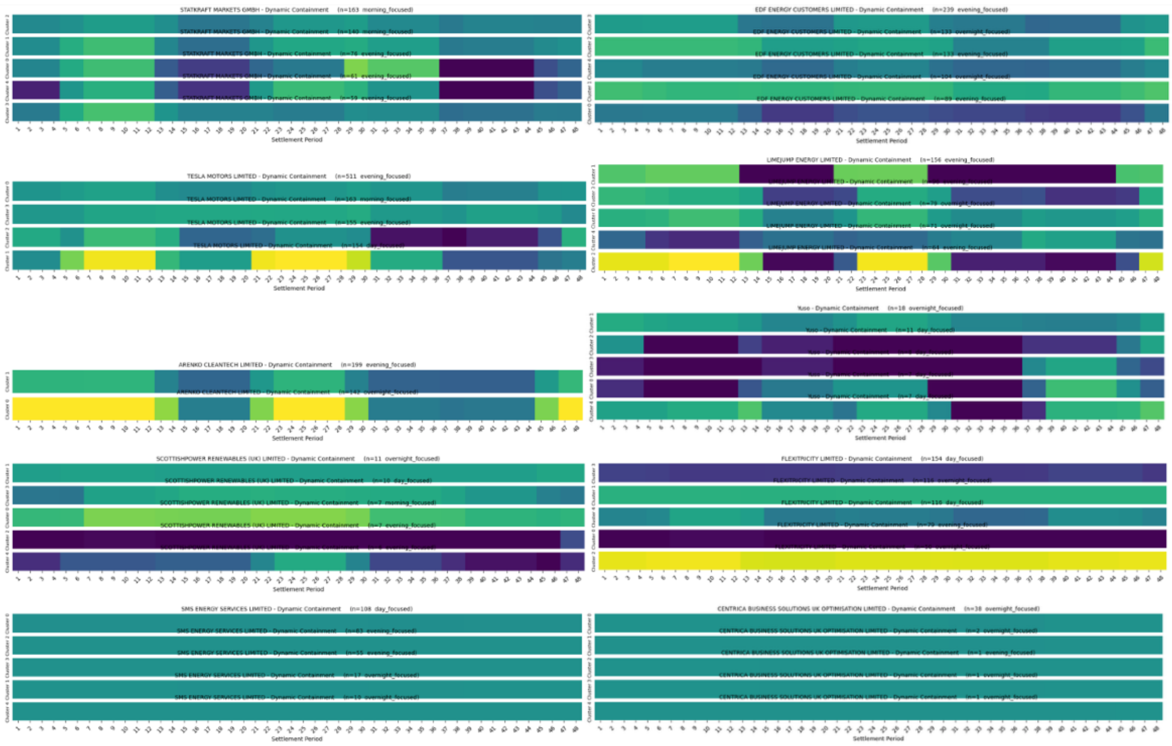

Figure 28: Heatmaps of Dynamic Containment bidding patterns by companies.

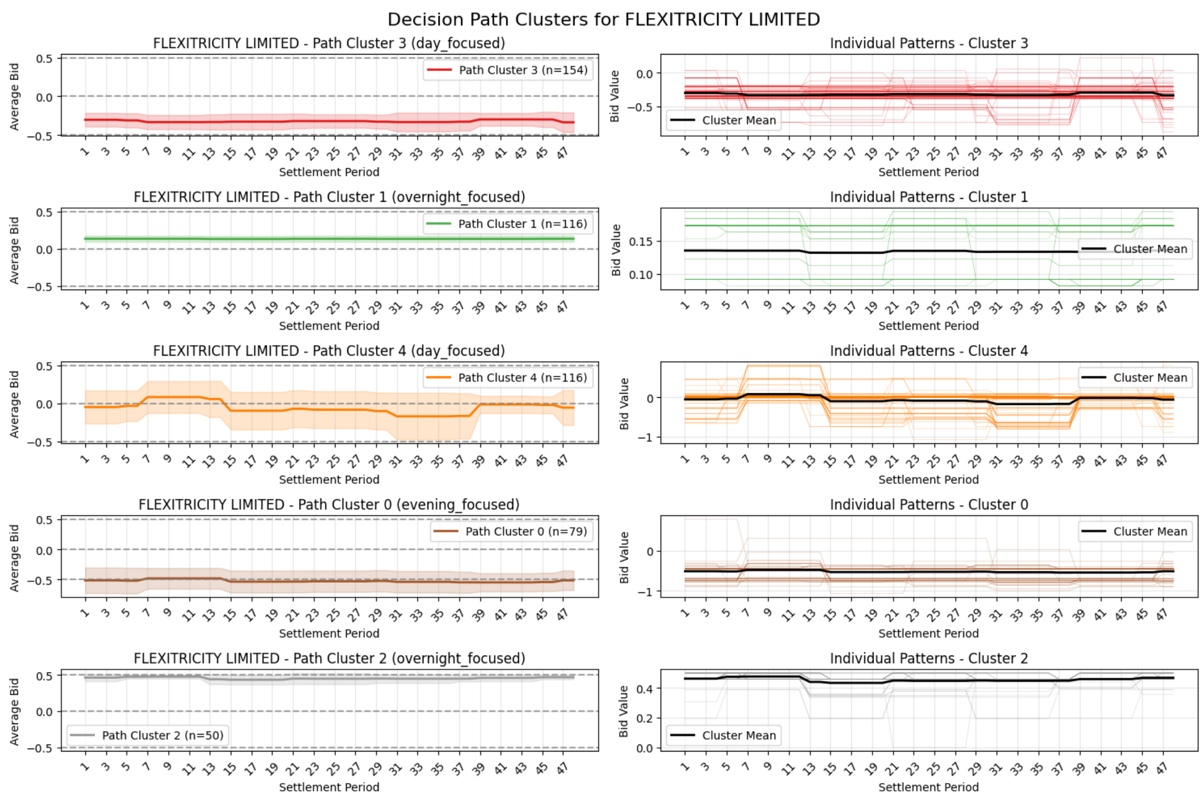


Figure 29: Flexitricity Limited bidding patterns in Dynamic Containment (example of asset-specific strategy).

### 3.5.2 Dynamic Moderation

DM shows similar peak-period positioning. EDF, Flexitricity and Statkraft maintain unidirectional strategies; Limejump exhibits contrasting positive/negative bidding, while Tesla balances participation moderately.

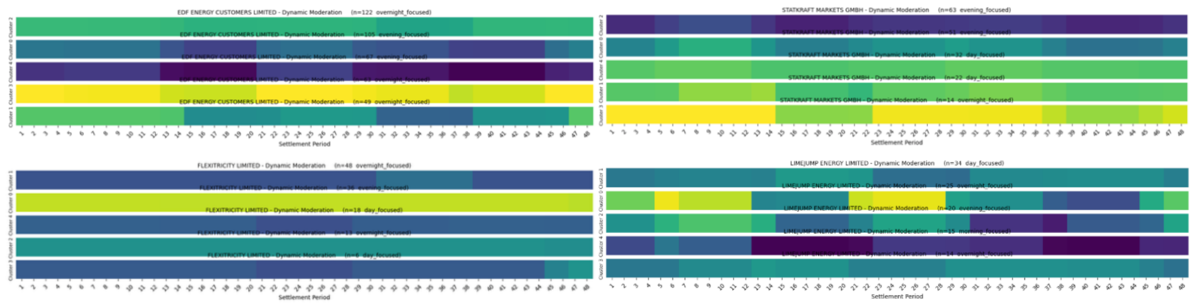

Figure 30: Heatmaps of Dynamic Moderation bidding patterns.

### 3.5.3 Dynamic Regulation

DR patterns are remarkably uniform across companies because of the stringent 60-minute minimum energy requirement favouring conservative volume approaches.

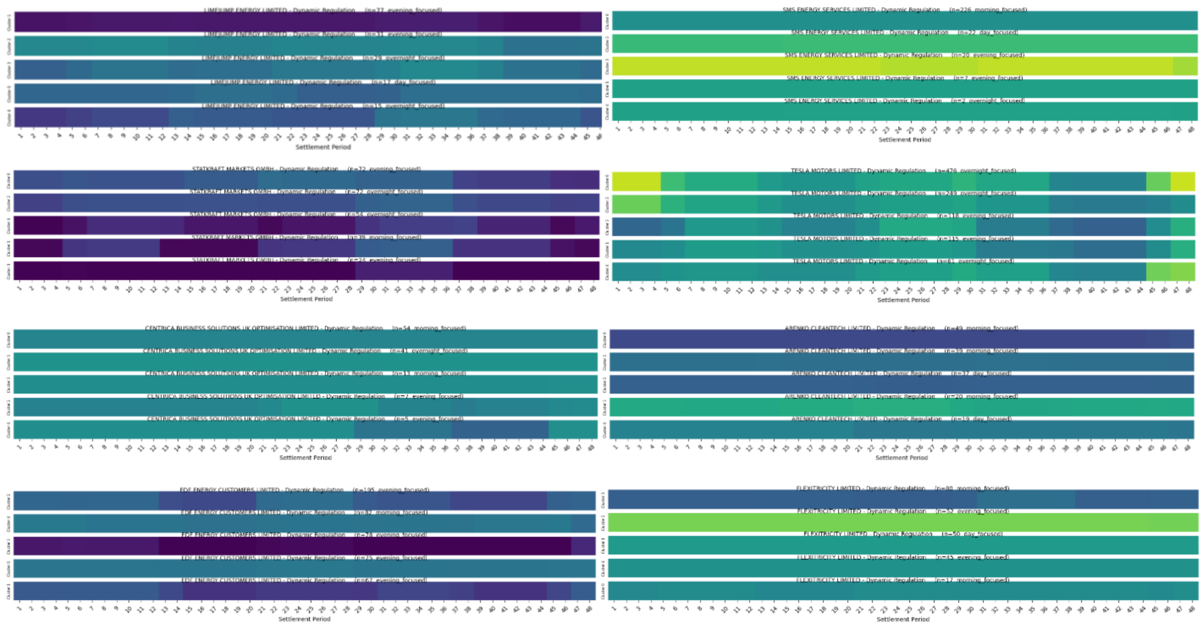

Figure 31: Heatmaps of Dynamic Regulation bidding patterns.

For Flexitricity, across Dynamic Containment, Dynamic Moderation and Dynamic Regulation, the association with Auction Unit remains consistently above 0.87, while the association with Month is notably lower. This implies that Flexitricity's bidding strategies are predominantly asset-driven rather than seasonally adjusted; each battery unit follows a customised strategy likely tailored to its operational characteristics or locational advantages, rather than applying a uniform portfolio-wide approach.

Table 4: Cramér's V between EAC market behavioural clusters and Auction Unit / Month for Flexitricity Limited

| | Cramér's V | | |
|---|---|---|---|
| | Dynamic Containment | Dynamic Moderation | Dynamic Regulation |
| Auction Unit | 0.8688 | 0.9115 | 0.8710 |
| Month | 0.2668 | 0.5293 | 0.2782 |

The contrasting temporal behaviours observed in Dynamic Containment and Dynamic Regulation can be directly linked to their distinct technical requirements (Table 5). Dynamic Containment's short 15-minute duration and post-fault activation allow operators to aggressively reposition their assets between charging and discharging across settlement periods, resulting in the pronounced intra-day shifts (e.g. noon charging, morning discharge) that align with EFA block price dynamics. In contrast, Dynamic Regulation mandates continuous provision for 60 minutes at full power over $\pm 0.2\,\text{Hz}$; this forces operators to maintain a stable state of charge and a conservative, almost unidirectional bidding profile to avoid energy depletion or saturation within the tendered hour.

Table 5: EAC service technical requirements overview

| Service | Duration Capability | Response Time | Frequency Range | Service Type |
|---|---|---|---|---|
| Dynamic Containment | 15 minutes | Sub-second | $> \pm 0.2\,\text{Hz}$ | Post-fault |
| Dynamic Moderation | 30 minutes | Fast-acting | $\pm 0.1$–$0.2\,\text{Hz}$ | Pre-fault (volatile) |
| Dynamic Regulation | 60 minutes | Proportional | Full power at $\pm 0.2\,\text{Hz}$ | Pre-fault (continuous) |
| Quick Reserve | 5 minutes | 1 min to full delivery | N/A | Reserve service |

### 3.5.4 Quick Reserve

QR strategies diversify significantly. Statkraft bids for charging at night and discharge during daytime; J Aron & Company shows pronounced discharge; Limejump focuses on reserve provision, and Arenko CleanTech exhibits large volume bids.

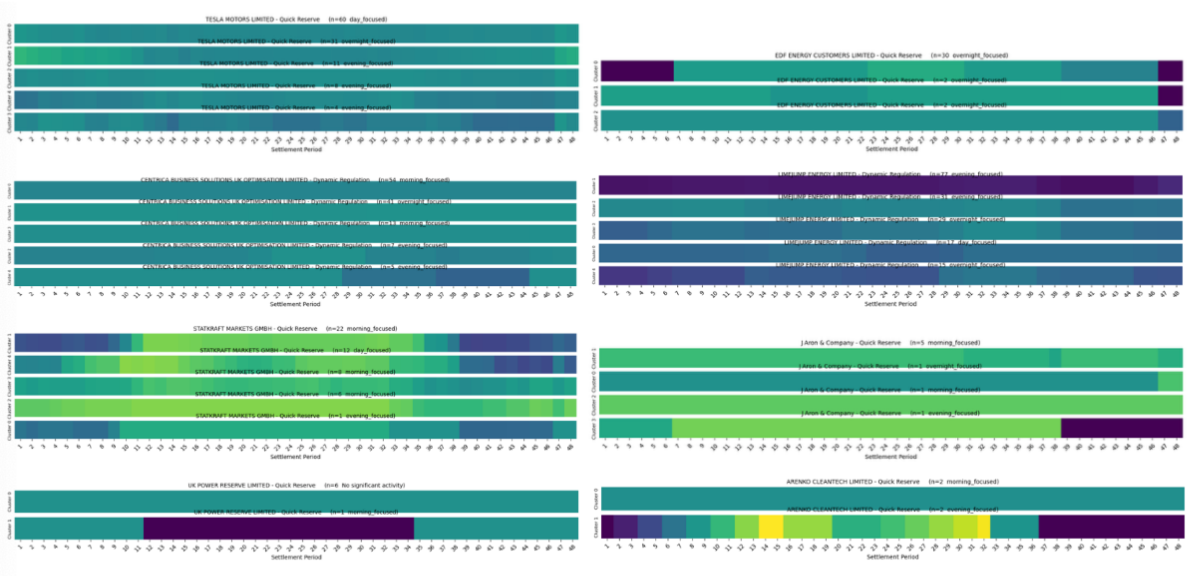

Figure 32: Quick Reserve heatmaps by company.

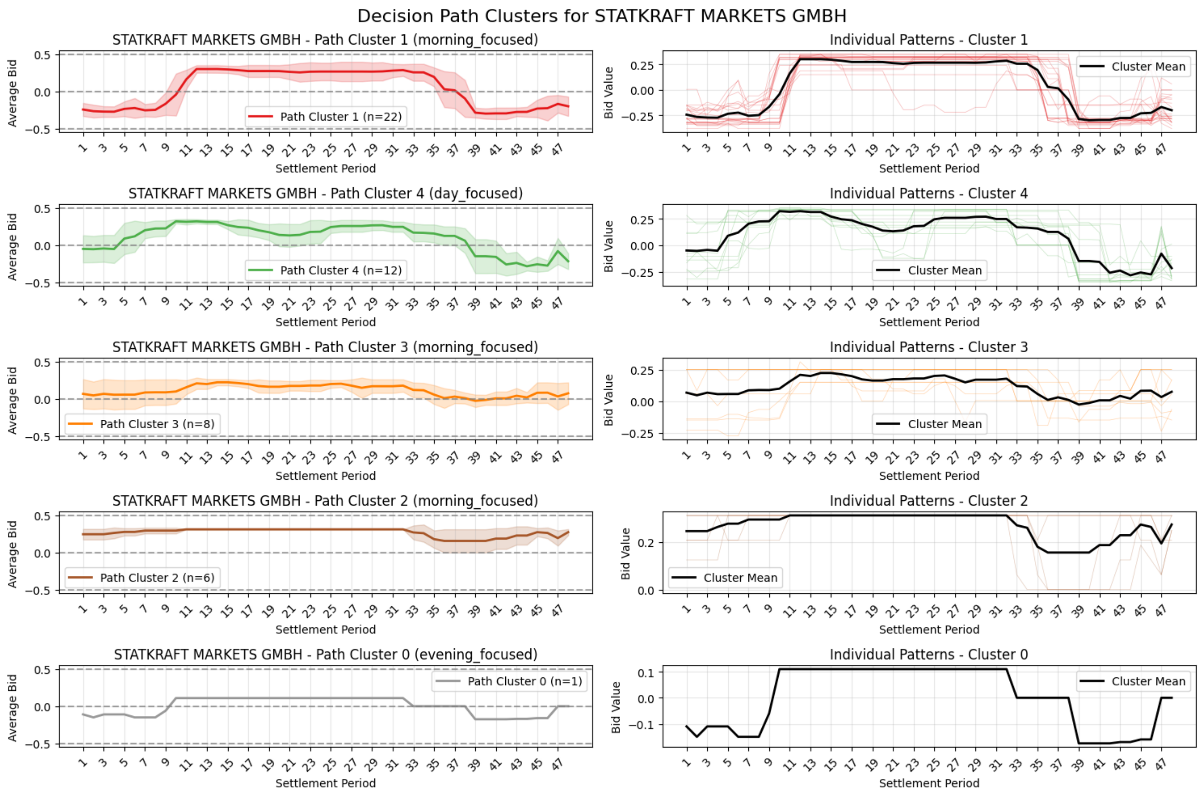


Figure 33: Statkraft Markets GMBH bidding patterns in Quick Reserve (example of asset-specific patterns).

### 3.5.5 Physical Notification (Wholesale)

PN patterns indicate peak-hour discharge consistent with energy arbitrage, with more bounded and volatile profiles compared to ancillary services. EDF shows discrete, volatile bidding across periods.

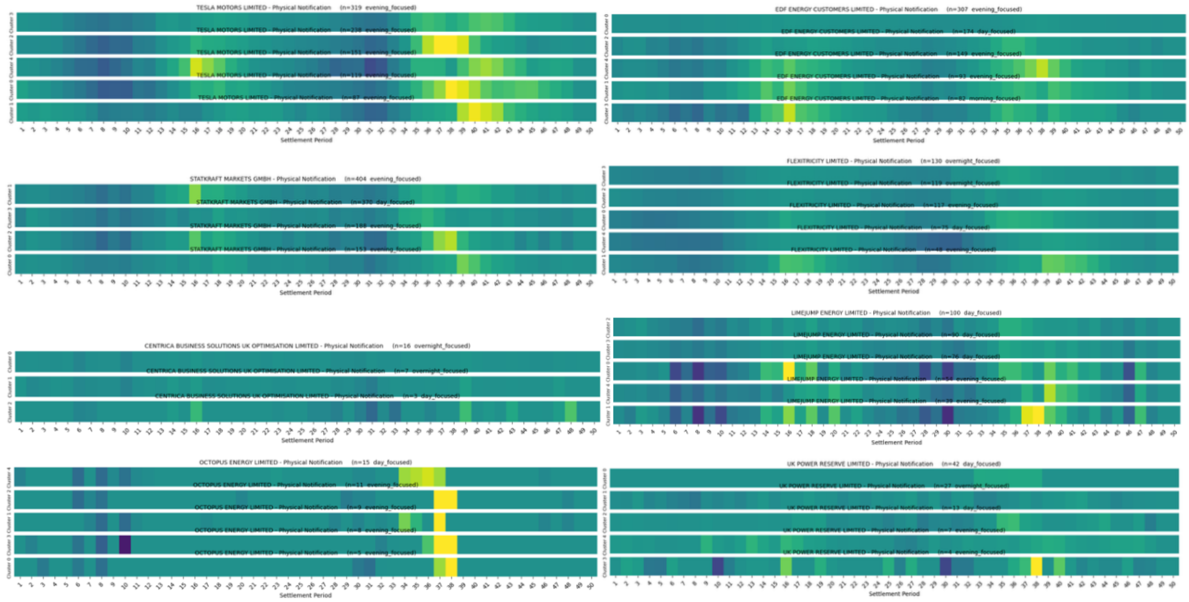

Figure 34: Physical Notification heatmaps.

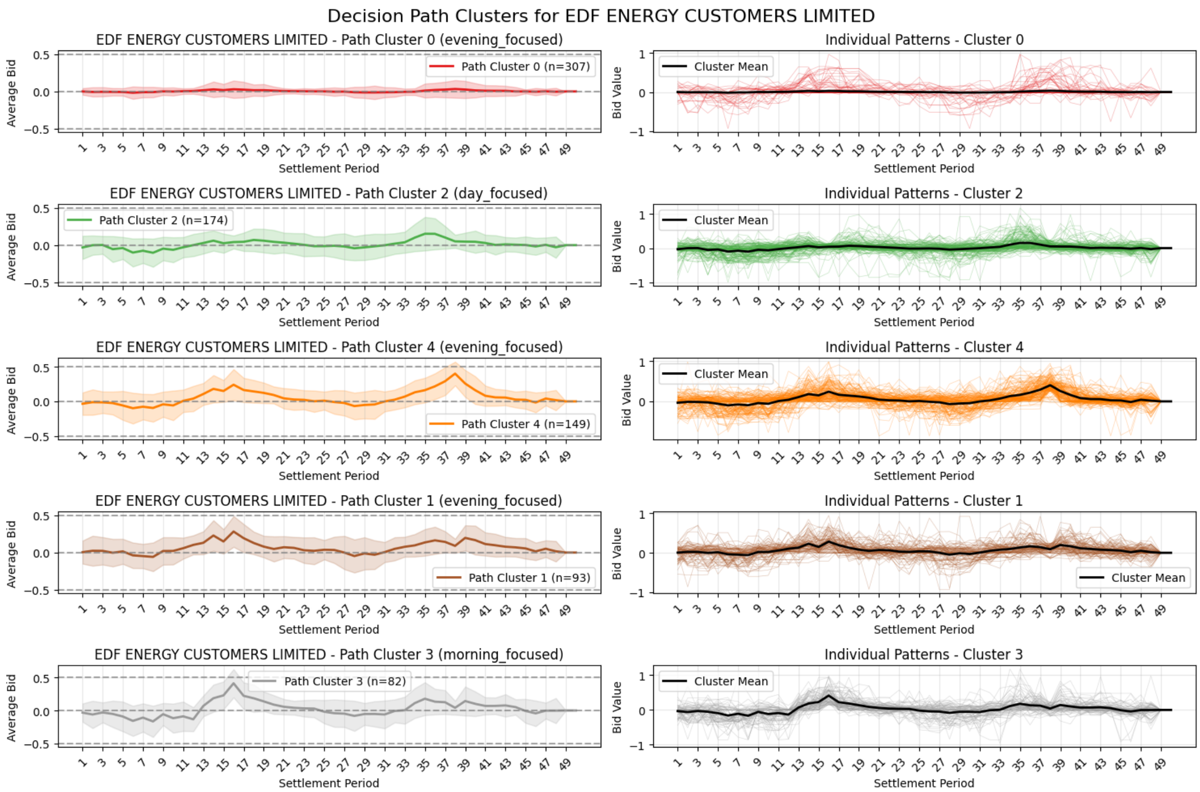


Figure 35: EDF Energy Customers Limited bidding patterns in Physical Notification (example of asset-specific patterns).

### 3.5.6 Balancing Mechanism

In the BM, companies like Centrica and Yuso demonstrate peak-period charging or generation reduction, contrasting with wholesale patterns. Statkraft maintains conventional generation increases; Limejump exhibits heavy unidirectional charging.

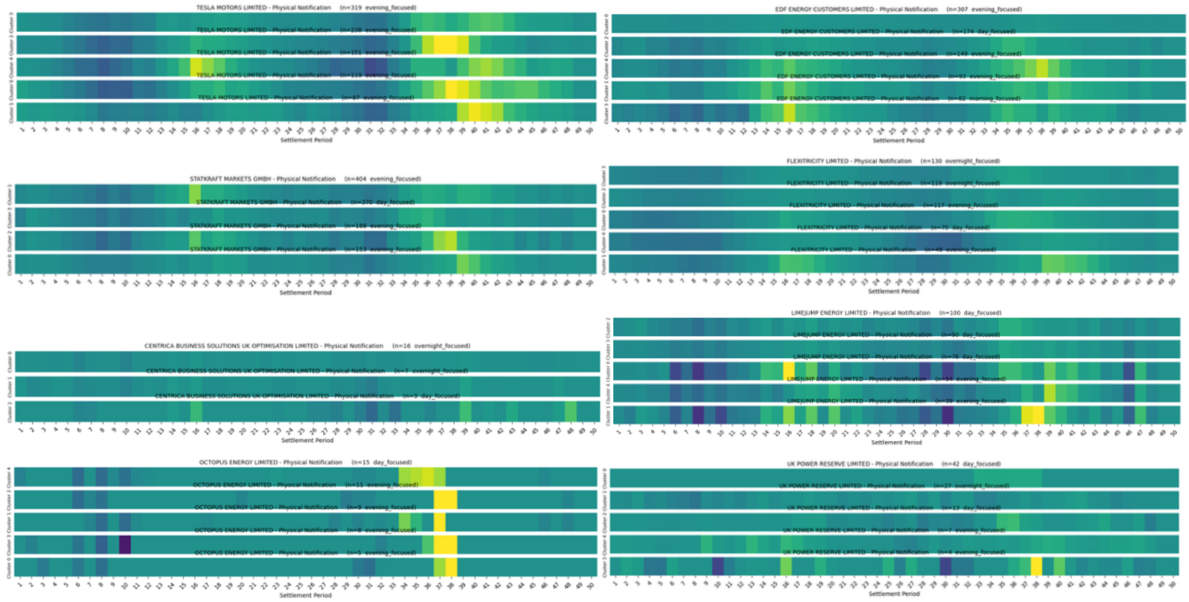

Figure 36: Balancing Mechanism heatmaps.

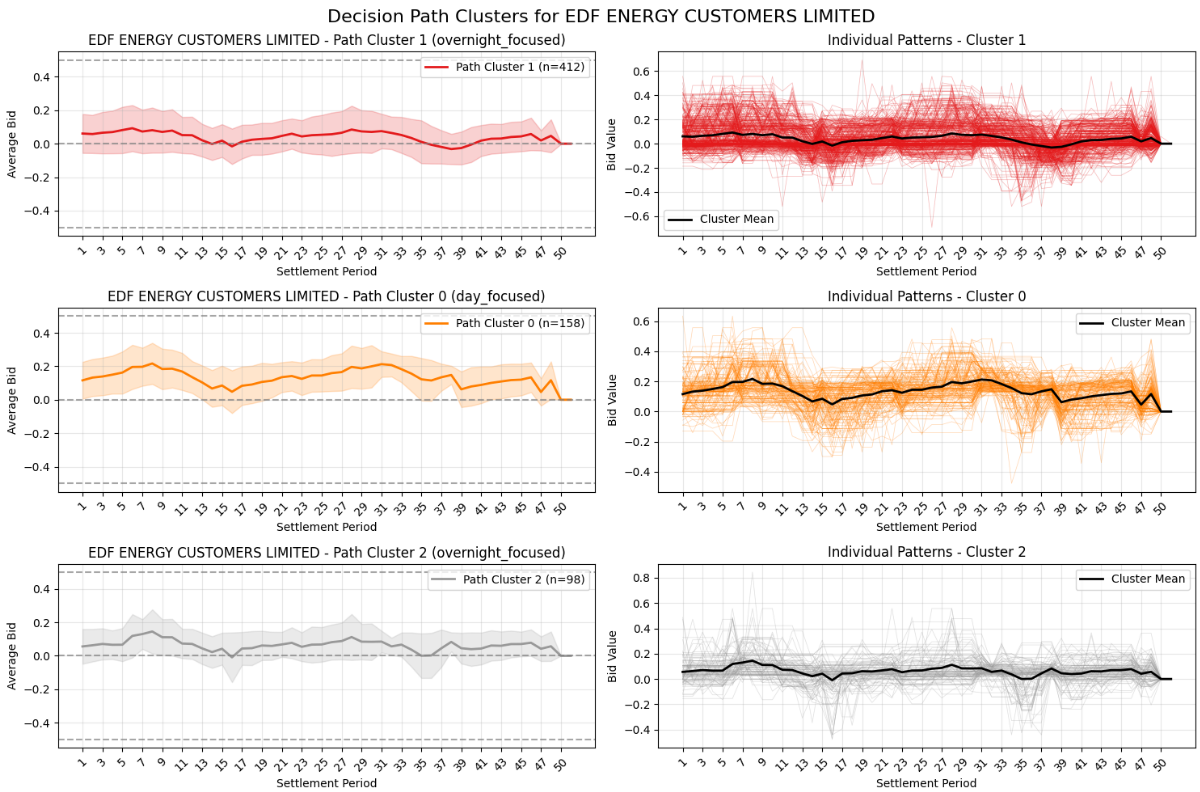


Figure 37: EDF Energy Customers Limited bidding patterns in Balancing Mechanism (example of asset-specific patterns).

### 3.5.7 Balancing Reserve

BR patterns feature gradual volume ramps and marked temporal differentiation. EDF's discrete bidding behaviour contrasts with other participants, possibly reflecting portfolio management to minimise penalties.

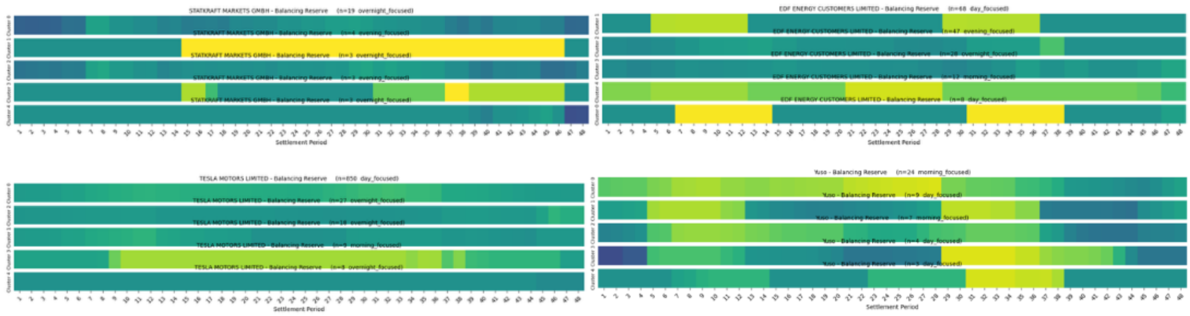

Figure 38: Balancing Reserve heatmaps.

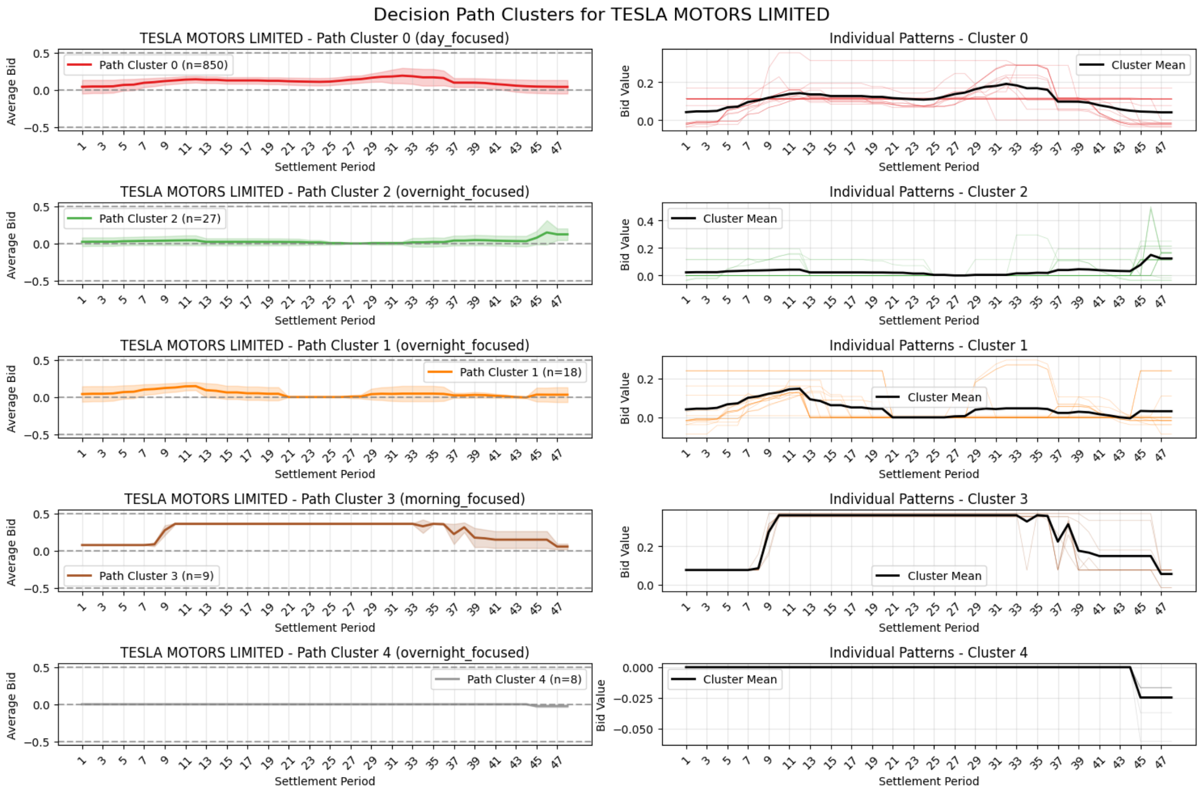


Figure 39: Tesla Motors Limited bidding patterns in Balancing Reserve (example of asset-specific patterns).

## 3.6 Behaviour Pattern Comparison by Companies

Rather than analysing bidding behaviours in isolated markets, the algorithm tracked the cluster assignments of all battery units simultaneously across market segments. The figure below (Figure 43) show consistant groups which auction units always appears in same sub-clusters.

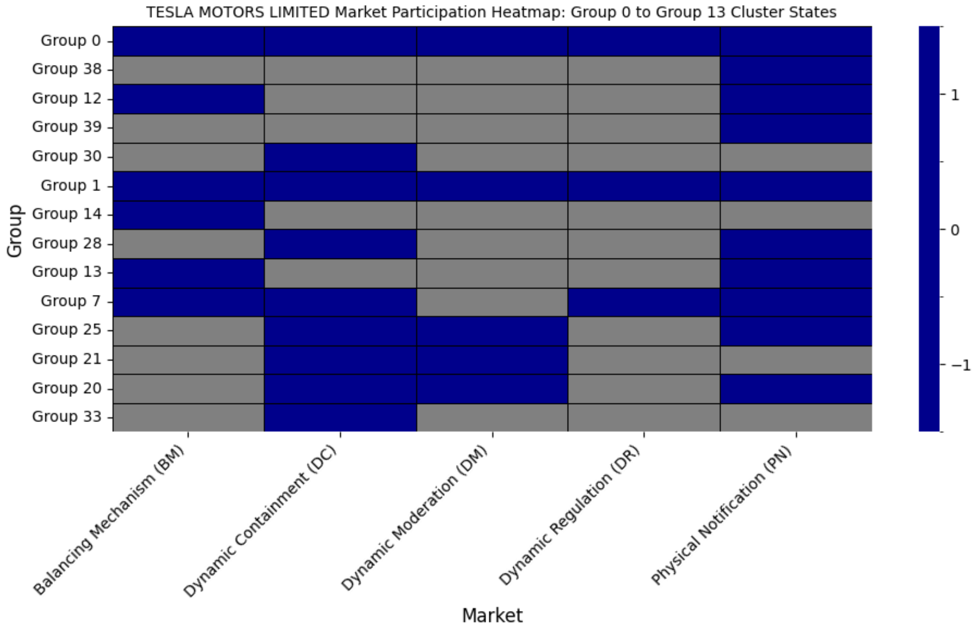


Figure 40: Tesla Motors Limited cross-market bidding patterns consistent groups.

**Tesla** uses Dynamic Containment/Moderation/Regulation actively for charging/SOC management and relies on Physical Notification for evening peak discharge; Balancing Reserve plays a supporting role.

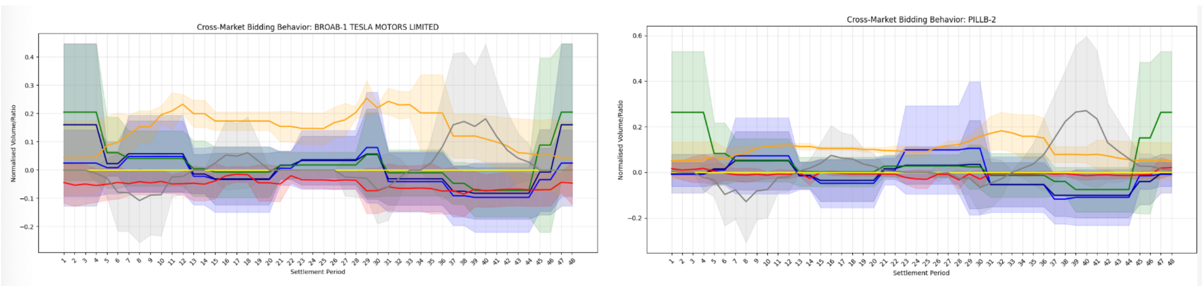


Figure 41: Tesla Motors Limited: examples of subplot of individual battery asset bidding patterns across markets. All Tesla assets display highly consistent bidding behaviours, reflecting the company's uniform fleet-wide strategy.

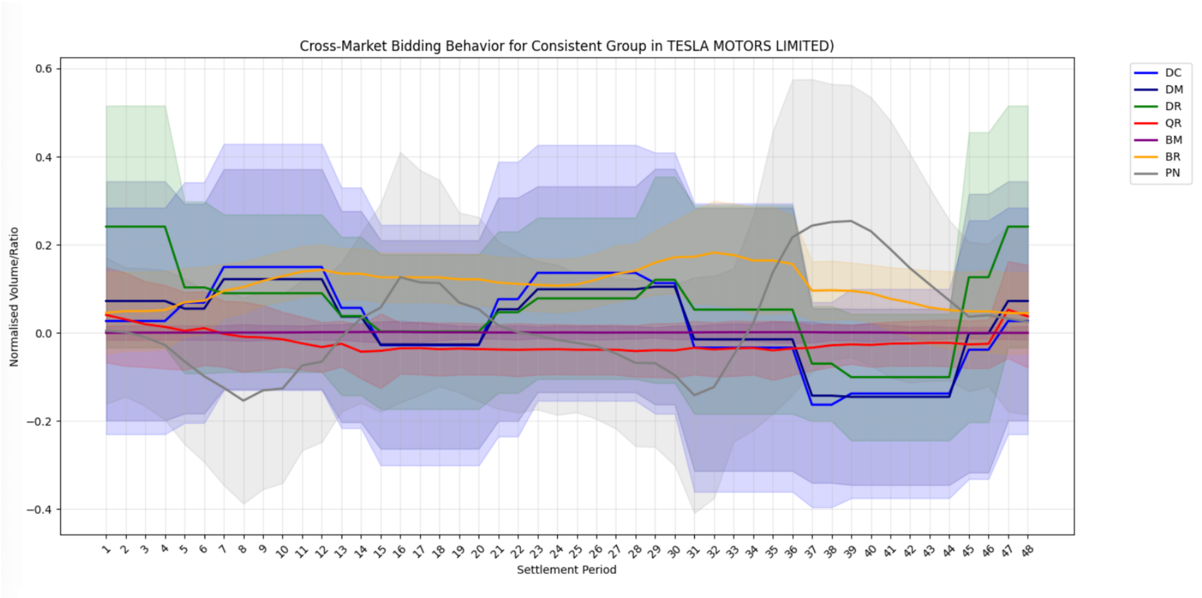


Figure 42: Tesla Motors Limited Group 0: cross-market bidding patterns. Tesla uses DC/DM/DR for charging and SOC management, PN for evening peak discharge, and BR as a supporting layer.

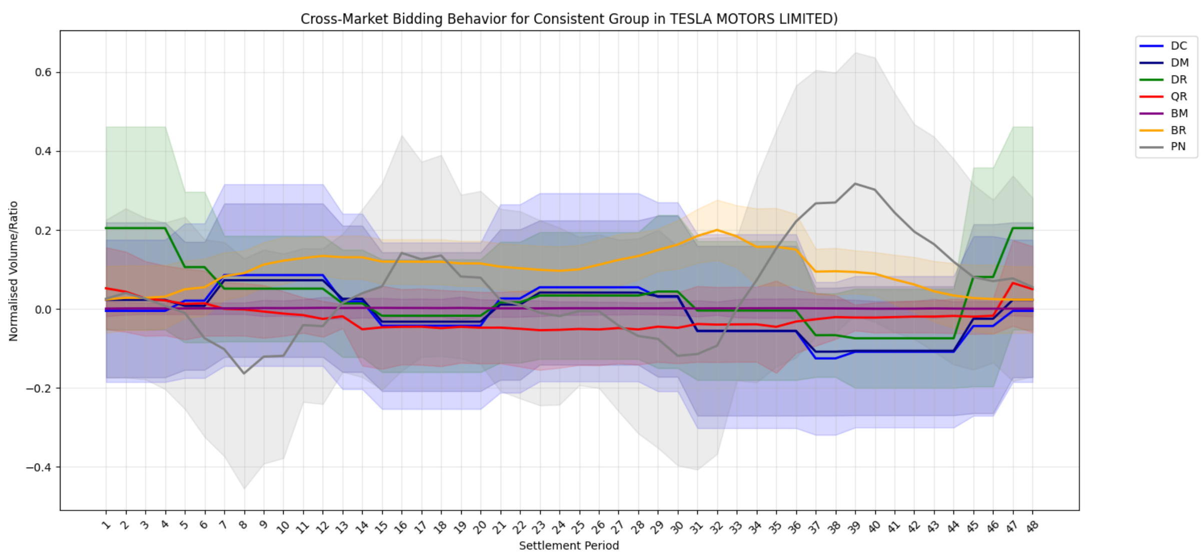


Figure 43: Tesla Motors Limited Group 1: cross-market bidding patterns.

Those two groups show distribution of monthly bids number diffrences. Group 1 (Figure 45)with tendency to filter later half of the year appears more conservative bidding pattern, with smaller standard diviation. The boxplot (Figure 46) shows there might capacity factors for the stacking strategy pattern.

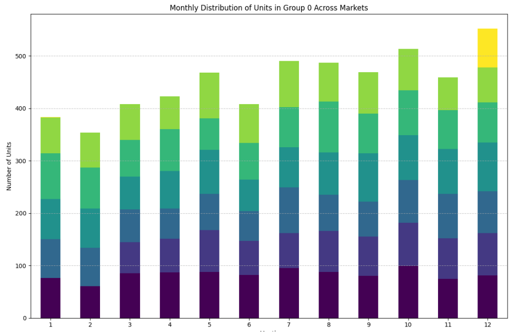


Figure 44: Tesla cross-market patterns group 0 month distribution.

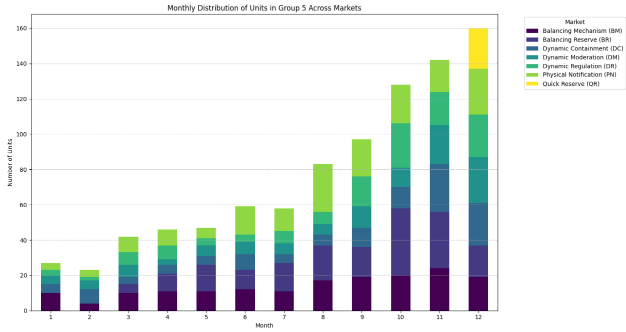


Figure 45: Tesla cross-market patterns group 1 month distribution.

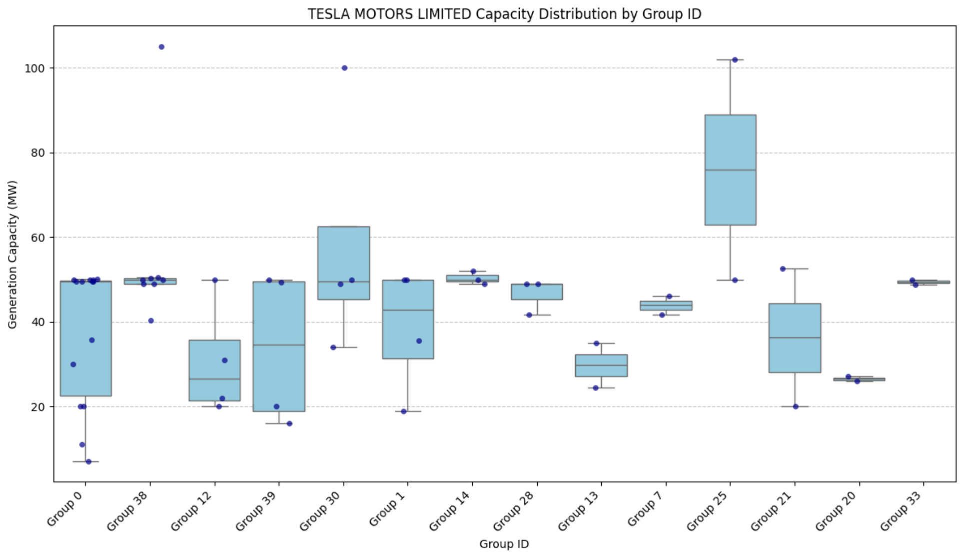


Figure 46: Tesla generation-capacity distribution by consistent bidding group.

**Statkraft** shows a QR/BR-led discharge strategy, with DR acting as a charging layer; PN is opportunistic around the evening peak.

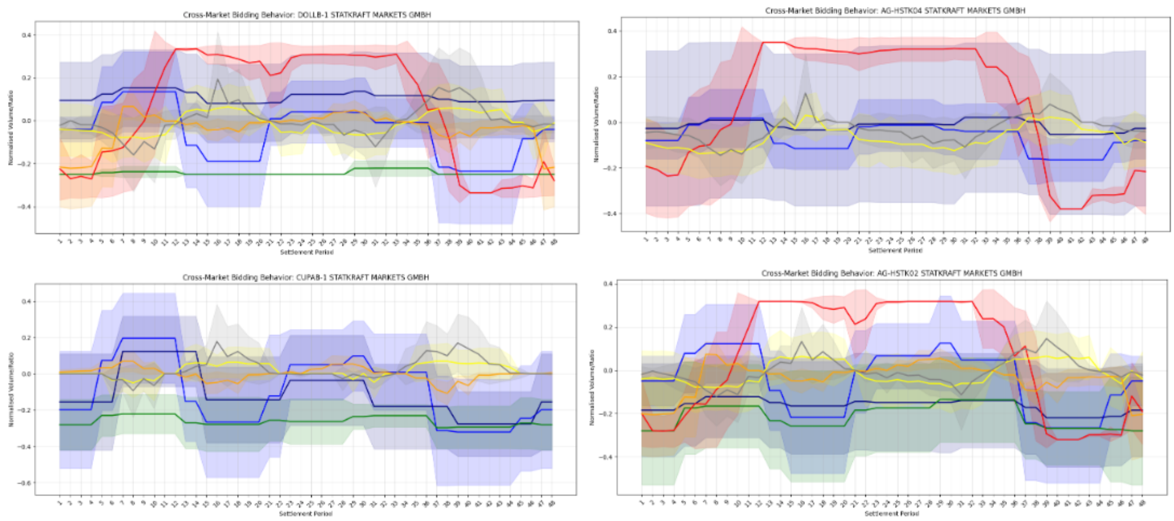


Figure 47: Statkraft Markets GmbH: cross-market bidding patterns by auction units

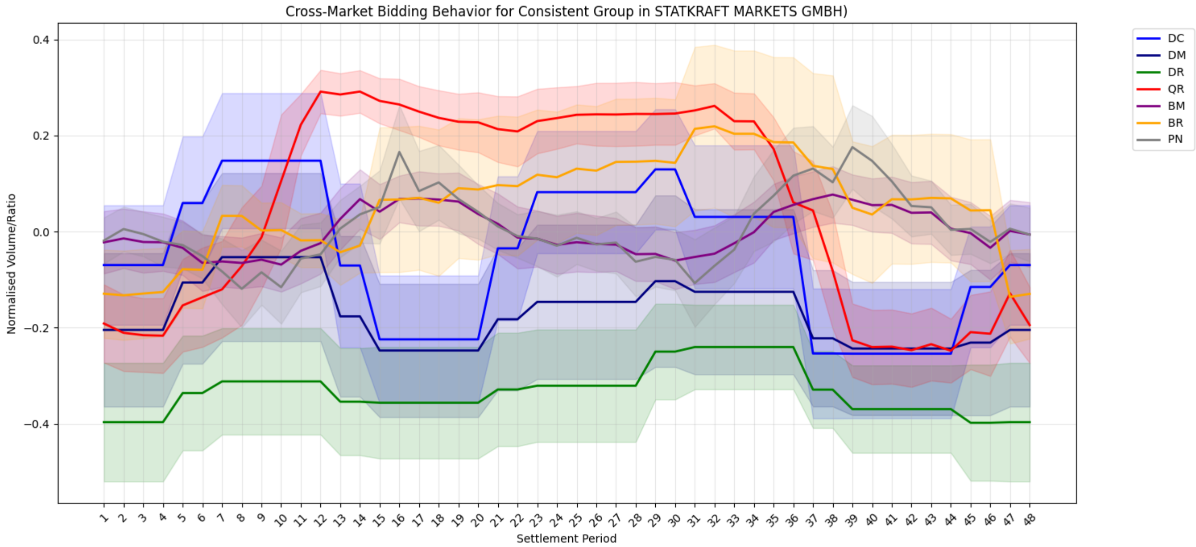


Figure 48: Statkraft Markets GmbH: cross-market bidding patterns. Discharge is led by Quick Reserve and Balancing Reserve, with DR serving as a charging layer and PN providing opportunistic evening peak discharge.

**EDF** is more BR/PN-led, with DR consistently negative and DC/DM providing complementary charging.

EDF Energy Customers consistent group: Two such groups within EDF Energy Customers Limited are particularly illustrative.

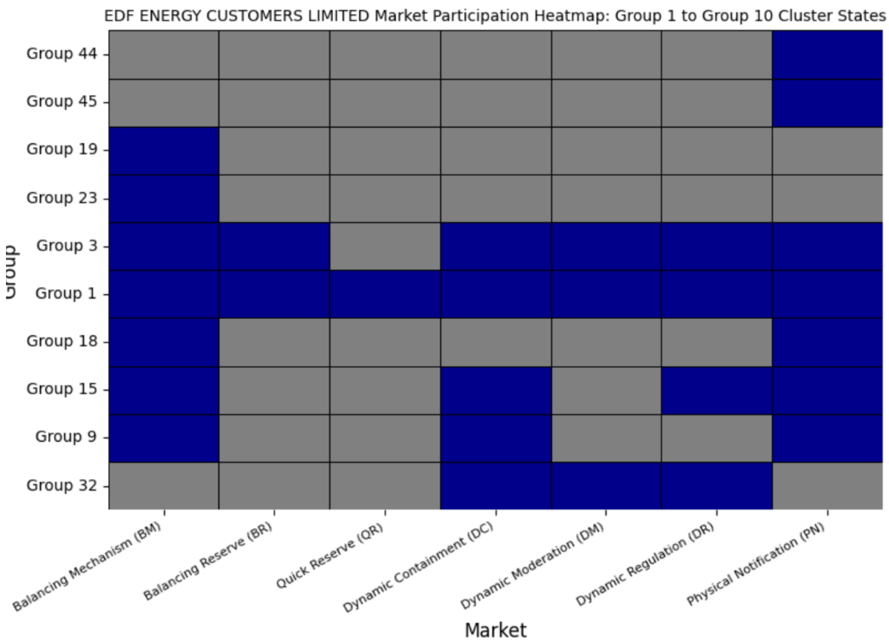


Figure 49: EDF cross-market bidding patterns consistent groups.

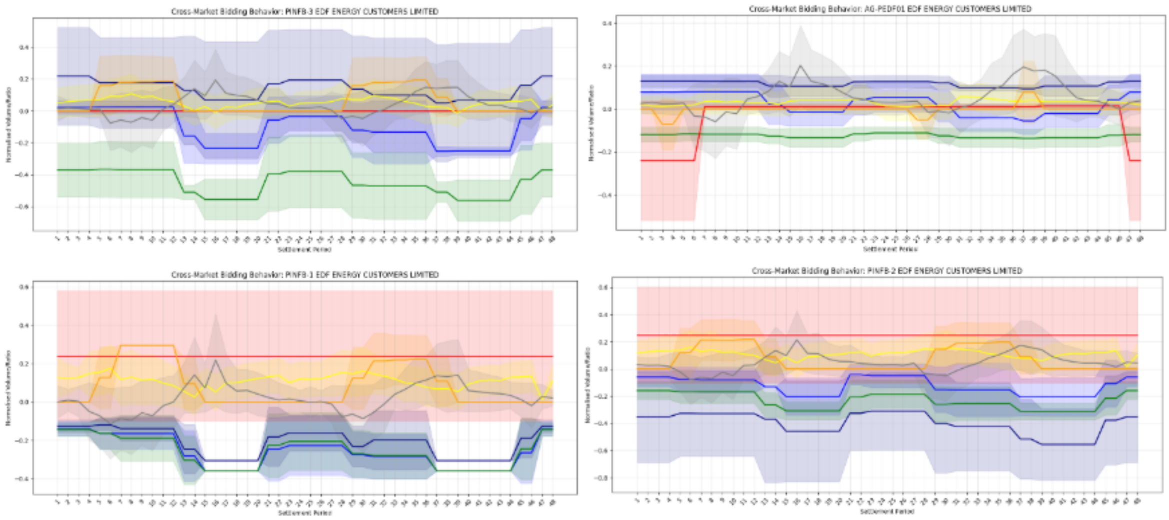

Figure 50: EDF Energy Customers Limited: cross-market bidding patterns by auction units.

**Group 1 EDF: BR-led strategy.** This cohort (Table **??**) is characterised by a strong Balancing Reserve presence. Asset COVNB-1, for instance, exhibits clearly positive BR blocks around SP6–12 and SP29–36, accompanied by a Physical Notification rise at SP35–40, while Dynamic Regulation remains consistently negative, serving as a charging.

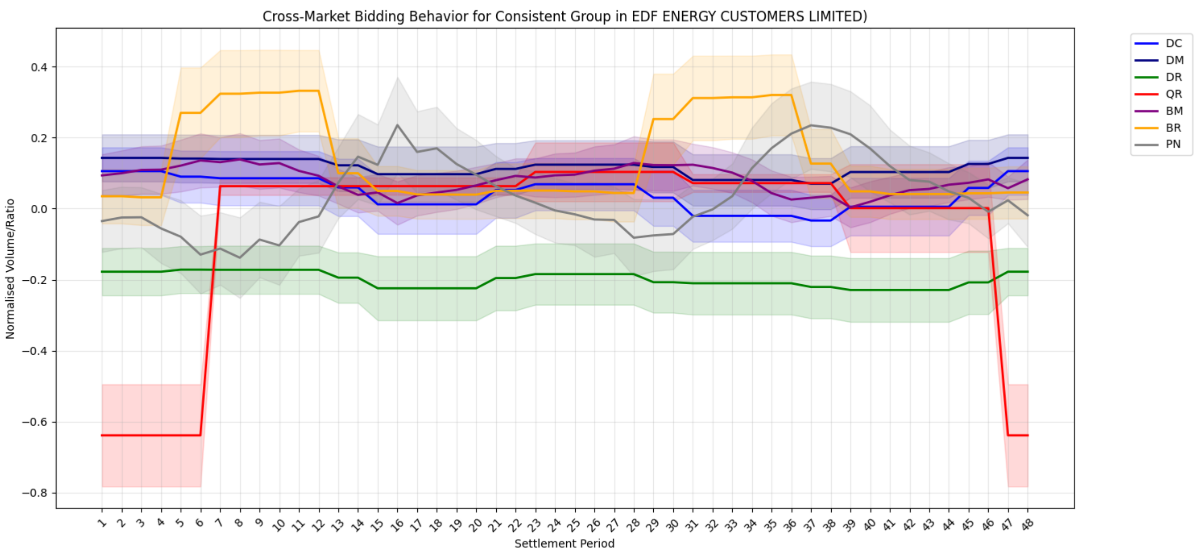


Figure 51: EDF Energy Customers Limited: cross-market bidding patterns. EDF relies on BR and PN for discharge, with QR in night and DR offering complementary charging.

**Group 3 EDF: DC/DM-driven dynamic-service strategy.** In contrast, Group 3 (Figur4e 52) assets such display positive DC/DM overnight and late day, turning negative during mid-day transition periods, and DR consistant charging. This group relies on frequency response services rather than balancing services for its revenue.

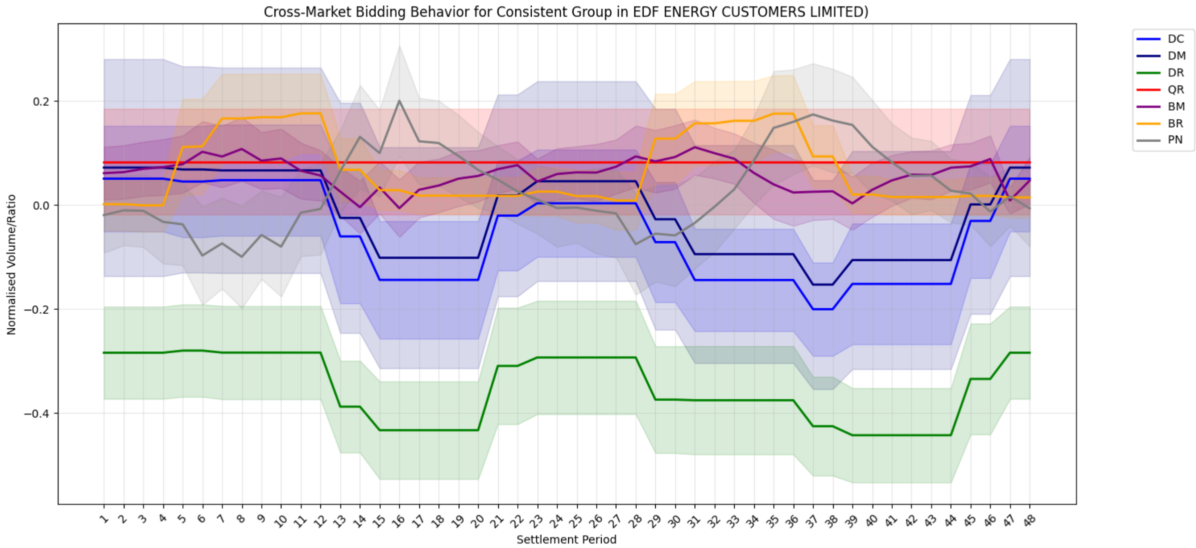


Figure 52: EDF Energy Customers Limited: cross-market bidding patterns. DR consistently negative.

The EDF boxplot shows that group capacity profiles differ: This suggests that asset size may partly influence group membership, although the evidence is limited because several groups contain only a small number of observations.

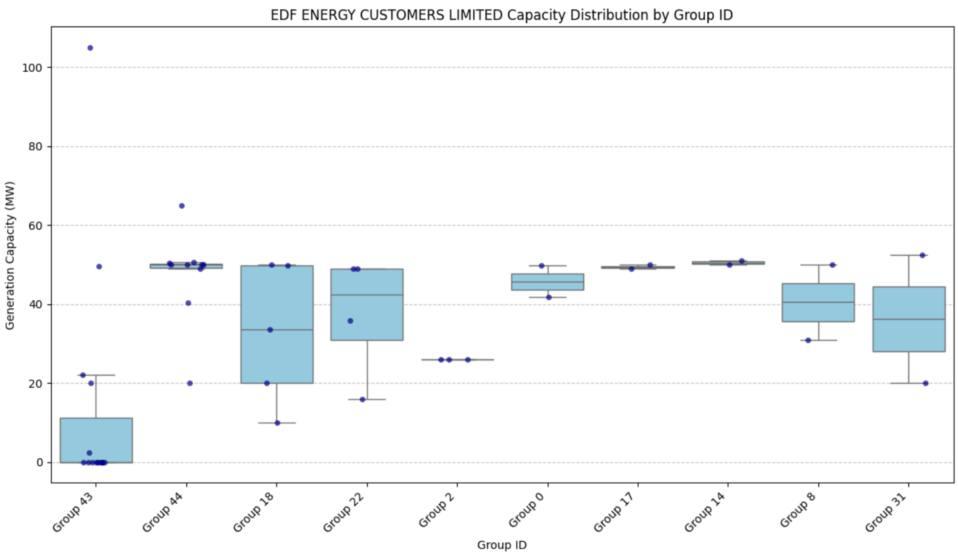


Figure 53: EDF generation-capacity distribution by consistent bidding group.

**Limejump** displays high-volume dual-directional bidding, while **Flexitricity** demonstrates asset-specific specialisation across frequency response services, found in high Cramér's V values for auction units in all Dynamic services (Table 4).

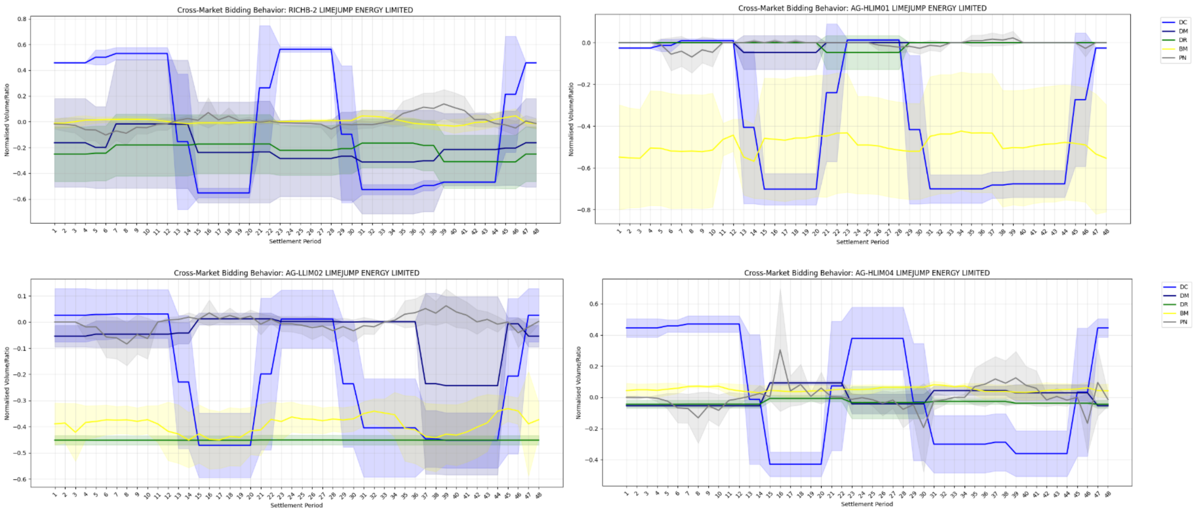


Figure 54: Limejump Energy Limited: cross-market bidding patterns.

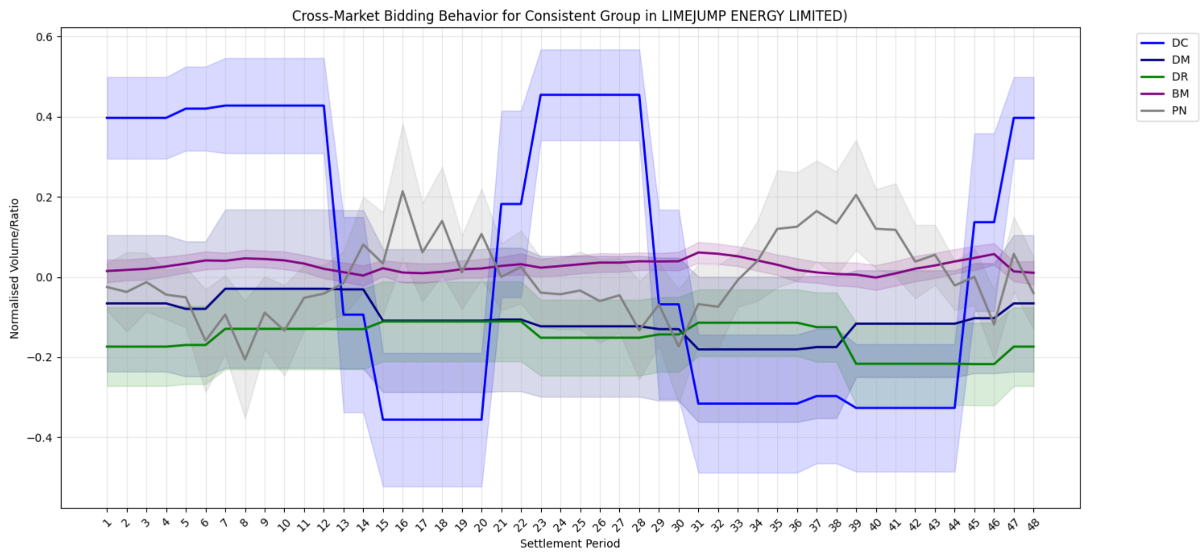


Figure 55: Limejump Energy Limited: cross-market bidding patterns.

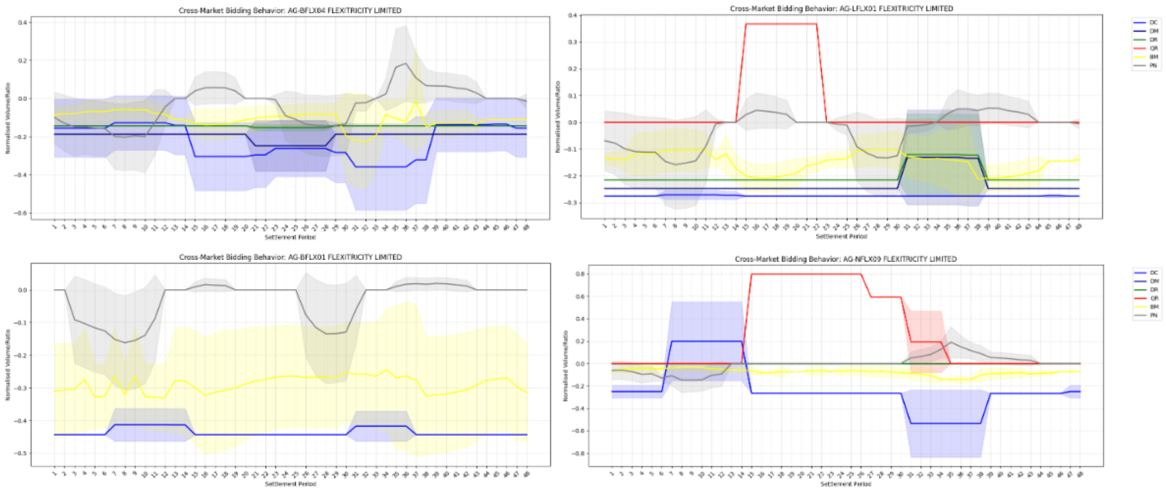

Figure 56: Flexitricity Limited: cross-market bidding patterns.

# 4 Discussion

The results demonstrate that machine learning classification can successfully identify distinct strategic fingerprints among UK battery storage operators. Random Forest achieves high macro F1-scores across markets, confirming that operator behaviours involve complex, non-linear interactions that Naive Bayes cannot capture.

**Research Question 1**: The high classification accuracy of Random Forest, especially in Dynamic Regulation (Macro F1 = 0.93) and Quick Reserve (0.87), provides strong evidence that battery operators possess unique, distinguishable bidding patterns. The drastic performance gap between Random Forest and Naive Bayes highlights that strategic differentiation relies on coordinated temporal patterns rather than isolated settlement-period decisions.

**Research Question 2**: Several distinctive patterns emerge. Dynamic Containment and Moderation strategies align closely with EFA block structures, with peak charging during morning and evening price peaks. Dynamic Regulation forces strategy convergence due to the 60-minute energy requirement, limiting differentiation. Quick Reserve, as a new market, already shows diverse adoption: Statkraft's day/night split, J Aron's discharge dominance, and Arenko's large-volume bids. In the Balancing Mechanism, some operators (Centrica, Yuso) charge during peak periods, opposing wholesale arbitrage behaviour, reflecting the real-time balancing function.

**Research Question 3**: Cross-market stacking reveals clear strategic archetypes. Tesla adopts a balanced portfolio, distributing charging/discharging across multiple services with a distinct evening peak discharge via PN. Statkraft and EDF concentrate discharge in BM, BR and QR, using DR primarily for charging. Limejump employs high-volume contrasting bids, suggesting an opportunity-focused approach. Flexitricity's highly asset-specific strategies indicate a flexibility-management business model that tailors each asset to local market conditions. These archetypes suggest that multiple viable business models coexist in the UK battery market.

## 5 Conclusions

This study successfully applies machine learning classification to UK battery storage bidding behaviours, demonstrating that operators possess distinct strategic fingerprints. Key findings include:

- Random Forest classifiers achieve high performance (Macro F1 up to 0.93) in identifying operators, confirming strategic diversity.
- Strategic patterns align with market structures: EFA blocks for DC/DM, regulatory constraints for DR, and distinct discharge/charge roles across BM, BR and wholesale.
- Four strategic archetypes analysed: portfolio diversification (Tesla), concentrated peak discharge (Statkraft/EDF), high-volume contrasting bids (Limejump), and asset-specific flexibility (Flexitricity).
- Feature importance reveals that competitive differentiation peaks during high-value settlement periods.

**Methodological contributions** include the first multi-market machine learning analysis of UK BESS bidding and the development of a decision path clustering technique to uncover tactical sub-patterns.

**Limitations** include the focus on BMU-registered assets only, potential misclassification of battery fuel types, reliance on PN as a wholesale proxy, the exclusion of Capacity Market data, and the single-year study period that may not generalise to other years. Future research should incorporate revenue outcomes, geographic and asset characteristics, more granular wholesale data, and longer time horizons to assess strategic evolution.

## Acknowledgements

The author acknowledges the Energy Institute, University College London, and Professor Aiden O'sullivan for supporting this research.

## References

[1] EIA (2024). *U.S. Energy Information Administration – Independent Statistics and Analysis*. Available at: `https://www.eia.gov/analysis/studies/electricity/batterystorage/pdf/battery_storage_2021`.

[2] Martins, J. and Miles, J. (2021). A techno-economic assessment of battery business models in the UK electricity market. *Energy Policy*, 148, p.111938. doi:10.1016/j.enpol.2020.111938.

[3] Bush, J. (2024). *GB BESS Outlook Q2 2024: Battery revenue stacking and dispatch optimization*. Modo Energy.

[4] Jennings, Z. (2025a). *GB BESS: Why top-performing batteries earn 30% higher revenues*. Modo Energy.

[5] Jennings, Z. (2025b). *GB Battery energy storage revenues hit a two-year high in December 2024*. Modo Energy.

[6] Modo Energy (2024). *GB Methodology – Indices*. Available at: `https://modoenergy.com/benchmarking/methodology/gb`.

[7] Elexon (2025). *BSCP 15: BM Unit Registration – Elexon Digital BSC*. Available at: `https://bscdocs.elexon.co.uk/bsc-procedures/bscp-15-bm-unit-registration`.

[8] NESO (2025). *Enduring Auction Capability (EAC)*. Available at: `https://www.neso.energy/industry-information/balancing-services/enduring-auction-capability-eac`.

[9] Chen, Q. et al. (2022). The competition and equilibrium in power markets under decarbonization and decentralization. *iEnergy*, 1(2), pp.188–203. doi:10.23919/ien.2022.0025.

[10] Merten, M. et al. (2020). Bidding strategy for battery storage systems in the secondary control reserve market. *Applied Energy*, 268, p.114951. doi:10.1016/j.apenergy.2020.114951.

[11] Rajah, I. et al. (2024). Degradation-Aware Derating of Lithium-Ion Battery Energy Storage Systems in the UK Power Market. *Electronics*, 13(19), 3817. doi:10.3390/electronics13193817.

[12] Fan, F., Nwobu, J. and Campos-Gaona, D. (2025). Co-Located Battery Energy Storage Optimisation for Dynamic Containment Under the UK Frequency Response Market Reforms. *CSEE JPES*. doi:10.17775/cseejpes.2023.01210.

[13] Aaltonen, H. et al. (2022). Bidding a Battery on Electricity Markets and Minimizing Battery Aging Costs: A Reinforcement Learning Approach. *Energies*, 15(14), 4960. doi:10.3390/en15144960.

[14] Seward, W., Qadrdan, M. and Jenkins, N. (2022). Revenue stacking for behind the meter battery storage. *Electric Power Systems Research*, 211, 108292. doi:10.1016/j.epsr.2022.108292.

[15] Tang, Q., Guo, H. and Chen, Q. (2022). Multi-Market Bidding Behavior Analysis of Energy Storage System Based on Inverse Reinforcement Learning. *IEEE Trans. Power Syst.*, 37(6), pp.4819–4831. doi:10.1109/tpwrs.2022.3150518.

[16] Weaver, N. (2022). *The evolution of the GB battery energy storage revenue stack*. Modo Energy.

[17] Wendel Hortop (2023a). *Dynamic Containment Low: why have prices increased so much?*. Modo Energy.

[18] Wendel Hortop (2024). *Balancing Reserve: Are ESO concerns over battery performance justified?*. Modo Energy.